Macroeconomia Dinâmica

Notas de aula

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Prefácio

Estas notas de aula foram elaboradas para as disciplinas de Macroeconomia III da Belavista e para a eletiva de Macroeconomia Dinâmica do Insper. A ideia da montagem deste curso foi ser uma adaptação "suave" entre o conteúdo de macroeconomia da graduação e dos cursos de mestrado/doutorado. Principalmente, dar uma roupagem mais moderna do que se faz hoje em macroeconomia em termos de microfundamentos. O texto será apresentado em inglês já que as aulas da Belavista também são neste idioma.

As notas são baseadas fortemente em diversos livros mais completos como [8], [12], [13], e [20].

Agradeço enormemente ao João Meningue, meu monitor na Belavista na primeira vez que dei o curso. O início deste projeto não seria possível sem você.

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1 Introduction

These notes provide a structured overview of the main topics covered in Dynamic Macroeconomics, focusing on dynamic macroeconomic models and intertemporal decision-making. The material is based primarily on [12], but also draws on complementary references, such as [20], whenever additional perspectives or formal developments are useful.

These notes start by introducing intertemporal consumption and saving decisions, highlighting how households allocate resources over time in response to income, interest rates, and preferences.

This framework provides the foundation for more advanced dynamic models used throughout modern macroeconomics. Building on this intuition, the notes then present the overlapping-generations (OLG) model, which departs from the representative-agent framework and allows the analysis of demographic structure, capital accumulation, and public policies such as pension systems.

Subsequent sections focus on labor supply decisions and the labor–leisure trade-off, both in static and dynamic settings. These models clarify how wages, taxes, and transfers affect individual labor supply choices and aggregate labor market outcomes. The notes then turn to investment decisions, discussing present value, risk, Tobin's Q, and adjustment costs, which together explain how firms decide how much to invest in the presence of uncertainty and intertemporal trade-offs.

The general equilibrium framework integrates households' consumption–saving and labor–leisure choices with firms' production and investment decisions, allowing for a coherent analysis of how prices coordinate economic activity across markets and over time. Finally, the notes introduce the tools of optimal control and dynamic programming, which are essential for solving infinite-horizon macroeconomic models and for analyzing the dynamics of capital accumulation, consumption, and policy interventions.

Overall, the objective of these notes is to provide an accessible and coherent roadmap to the core theoretical models of modern macroeconomics, highlighting their economic intuition while maintaining the formal structure necessary for analytical and quantitative applications.

2 Consumption and Savings Model

2.1 The Keynesian View of Consumption

Keynes' view of consumption, as presented in Kurlat's book, is that consumption depends on current income through a consumption function:

$$C = c(Y), \tag{2.1}$$

where C is aggregate consumption, Y is the output (income) of the economy and $c(\cdot)$ is the consumption function. The derivative $c'(Y)$ is the marginal propensity to consume (MPC), the extra consumption for each extra unit of income, while $c(Y)/Y$ is the average propensity to consume (APC), the share of income that is consumed. Keynes stated that, when Y grows, C also grows, but by less, that is, $0 < c'(Y) < 1$. He also stated that the share of income consumed falls as income grows, that is, MPC $<$ APC. This is equivalent to saying that the elasticity of consumption with respect to income is positive but less than 1:

$$\frac{c'(Y)\,Y}{c(Y)} = \frac{d\log c(Y)}{d\log Y} = \frac{\text{MPC}}{\text{APC}} < 1. \tag{2.2}$$

2.2 A Two-Period Model of Consumption

Consider an economy composed of one family and two periods, t_1 and t_2. The utility function of this family (or this economy) is:

$$U(c_1, c_2) = u(c_1) + \beta u(c_2), \tag{2.3}$$

where c_1 and c_2 are the consumption of this family in periods 1 and 2, respectively. $0 < \beta < 1$ is the discount factor: the family prefers consumption in period 1 to consumption in period 2, that is, it is impatient. The saving is:

$$a = y_1 - c_1 \tag{2.4}$$

a in the second period will be boosted by the real interest rate r. This way,

$$c_2 = y_2 + (1 + r)a \tag{2.5}$$

rearranging,

$$c_1 + \frac{1}{1+r}c_2 = y_1 + \frac{1}{1+r}y_2 \tag{2.6}$$

Note that $\frac{1}{1+r}$ represents the value of c_2 and y_2 in period 1; thus, one can conclude that the interest rate indicates how much the market pays in terms of c_2 consumption for less consumption in c_1. Therefore, a high interest rate means a higher cost of consumption in the first period. Still at this point, the income of the family in two periods is represented by $y_1 + \frac{1}{1+r}y_2$, which means the present value of all income.[1]

Analyzing the consumer optimization problem:

[1] Making it explicit, $\frac{1}{1+r}$ gives the value of future consumption or income in the current period.

$$\begin{aligned} \max_{c_1,\, c_2} \quad & u(c_1) + \beta\, u(c_2) \\ \text{s.t.} \quad & c_1 + \frac{1}{1+r}\, c_2 \leq y_1 + \frac{1}{1+r}\, y_2 \end{aligned} \tag{2.7}$$

This problem will look for the point at the budget constraint curve where the indifference curve of the trade between c_1 and c_2 is tangential to the budget constraint curve, as is shown in the following figure:

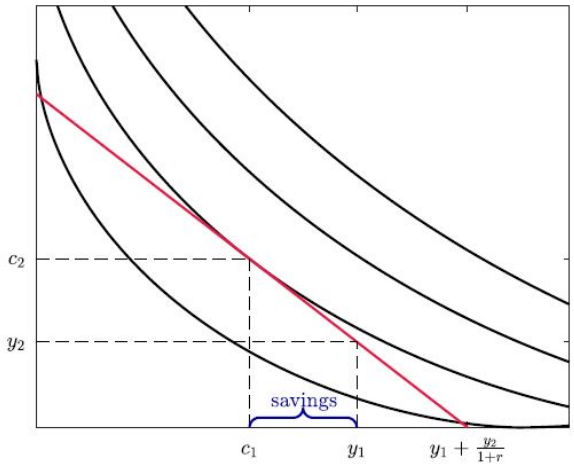


Figure 1: Source: [12]

Using the *Lagrangian* allows one to find the solution of the maximization through the first-order conditions.

$$\begin{aligned} \mathcal{L}(c_1, c_2, \lambda) &= u(c_1) + \beta\, u(c_2) - \lambda\Big[c_1 + \tfrac{1}{1+r} c_2 - y_1 - \tfrac{1}{1+r} y_2\Big] \\ \frac{\partial \mathcal{L}}{\partial c_1} &= u'(c_1) - \lambda = 0 \\ \frac{\partial \mathcal{L}}{\partial c_2} &= \beta\, u'(c_2) - \frac{\lambda}{1+r} = 0 \end{aligned}$$

Rearranging,

$$u'(c_1) = \beta(1+r)u'(c_2) \tag{2.8}$$

Equation (2.8) is the *Euler equation*. Note that this informs the trade-off that the household faces between changing consumption in the first period and consumption in the second period. If the consumption of one unit happens in the first period, there will be $u'(c_1)$ units of utility (left hand side of (2.8)).

However, suppose the family chooses in the second period. In that case, they will obtain $(1+r)$ units of the same consumption in the future (interest rate), which means $u'(c_2)$ marginal utility multiplied by β for the household's impatience. The maximization occurs when (2.8) holds. It is important to note that the household can obtain either positive or negative savings in the first period, depending on its preferences, price, or interest rate. The household aims to smooth its consumption.

2.2.1 The Effect of Interest Rate

At this point, it is perceptible that this model is very similar to consumer theory in microeconomics. Therefore, it is important to examine the substitution effect and the income effect that are at play.

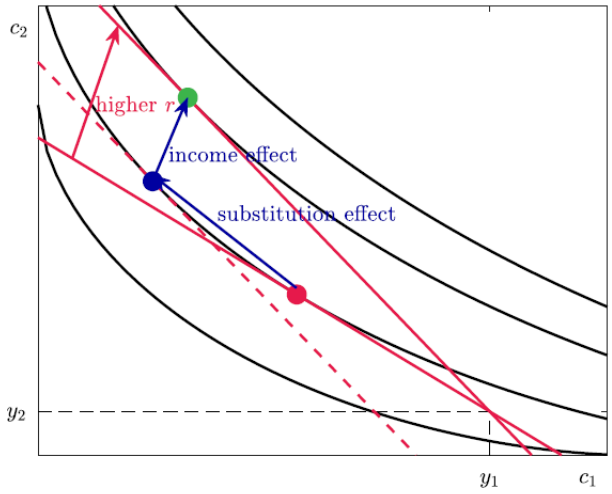


Figure 2: Source: [12]

Note that when the interest rate increases, there will be a new slope in the budget constraint curve. As a consequence, a new c_1, c_2 indifference curve will be tangent to the budget line, and a new composition of c_1 and c_2 will be chosen. A part of this choice is the consequence of the substitution effect, and another is the income effect, as Figure 2 shows.

The direction of the effects depends on the household's preferences regarding positive or negative savings. If the family has negative savings, the increase in interest rate will have a negative impact on the family's consumption, which means a lower indifference curve of c_1, c_2.

2.2.2 An Explicit Example

Assume the utility function is a CRRA, that is:

$$u(c) = \frac{c^{1-\sigma}}{1-\sigma} \tag{2.9}$$

The derivative is:

$$u'(c) = c^{-\sigma} \tag{2.10}$$

remembering (2.8), that is, $u'(c_1) = \beta(1+r)u'(c_2)$. Applying (2.10) into (2.8), results in

$$\begin{aligned} c_1^{-\sigma} &= \beta(1+r)c_2^{-\sigma} \\ \Rightarrow c_2 &= [\beta(1+r)]^{\frac{1}{\sigma}} c_1 \end{aligned} \tag{2.11}$$

using the budget restriction, $c_1 + \frac{1}{1+r}c_2 = y_1 + \frac{1}{1+r}y_2$, and applying (2.11),

$$c_1 + \frac{1}{1+r}[\beta(1+r)]^{\frac{1}{\sigma}} c_1 = y_1 + \frac{1}{1+r}y_2 \tag{2.12}$$

As a result, the optimal consumption of c_1 is given by:

$$c_1 = \frac{y_1 + \frac{1}{1+r} y_2}{1 + \beta^{\frac{1}{\sigma}} (1+r)^{\frac{1}{\sigma}-1}} \tag{2.13}$$

Note that c_1 optimal is expressed in terms of income in present value $y_1 + \frac{1}{1+r} y_2$ and on β, r and σ through the term $1 + \beta^{\frac{1}{\sigma}} (1+r)^{\frac{1}{\sigma}-1}$.

2.3 Extension to Many Periods

Sometimes it could be necessary to analyze the problem of consumption for more than two periods. To make it, the simple model's principles are used. Thus, the preferences of the household are now:

$$\sum_{t=0}^{T} \beta^t u(c_t) \tag{2.14}$$

Note that the consumption of future periods affects the utility, but the more distant this consumption is, the less impact it has on utility at time t, since $\beta < 1$. Consider $T = \infty$ as an assumption. Since a_t represents the savings of the household in time t, a_{t+1} is expressed by:

$$a_{t+1} = (1+r) a_t + y_t - c_t \tag{2.15}$$

If $T = \infty$, the family will face infinite constraints that relate two consecutive periods like (2.15). To solve the maximization problem, a simple constraint is necessary, which can be achieved by substituting one term for another with the savings, that is,

$$a_1 = y_0 - c_0 + (1+r) a_0 \tag{2.16}$$
$$a_2 = y_1 - c_1 + (1+r) a_1 \tag{2.17}$$

It is possible to substitute a_1 expression at a_2 expression,

$$a_2 = y_1 - c_1 + (1+r)[y_0 - c_0 + (1+r) a_0] \tag{2.18}$$

For a_3,

$$\begin{aligned} a_3 &= y_2 - c_2 + (1+r)[y_1 - c_1 + (1+r)[y_0 - c_0 + (1+r) a_0]] \\ a_3 &= y_2 - c_2 + (1+r)(y_1 - c_1) + (1+r)^2 (y_0 - c_0) + (1+r)^3 a_0 \end{aligned} \tag{2.19}$$

For period-T, the constraint is:

$$a_{T+1} = \sum_{t=0}^{T} y_t (1+r)^{T-t} - \sum_{t=0}^{T} c_t (1+r)^{T-t} + (1+r)^{T+1} a_0 \tag{2.20}$$

or

$$\frac{a_{T+1}}{(1+r)^T} + \sum_{t=0}^{T} \frac{c_t}{(1+r)^t} = \sum_{t=0}^{T} \frac{y_t}{(1+r)^t} + (1+r) a_0 \tag{2.21}$$

Equation (2.21) contains much information. The sum $\sum_t c_t/(1+r)^t$ is the present value of consumption in all periods, and $\sum_t y_t/(1+r)^t$ is the present value of income. The term $(1+r)a_0$ is the value of the assets the household is born with (it is zero in the two-period example, where $a_0 = 0$). Finally, $a_{T+1}/(1+r)^T$ is the present value of the assets left after the final period: if it is positive, the household leaves a bequest; if it is negative, it dies in debt. When $T \to \infty$, without any restriction the household could borrow forever and roll the debt over, consuming more than its resources (a Ponzi scheme). The *no-Ponzi* condition rules this out:

$$\lim_{T\to\infty} \frac{a_{T+1}}{(1+r)^T} \geq 0. \tag{2.22}$$

Moreover, as utility is increasing in consumption, a household that leaves positive assets in the limit could have consumed them, so at the optimum the inequality holds with equality (the transversality condition): $\lim_{T\to\infty} a_{T+1}/(1+r)^T = 0$. So (2.21) becomes

$$\sum_{t=0}^{\infty} \frac{c_t}{(1+r)^t} = \sum_{t=0}^{\infty} \frac{y_t}{(1+r)^t} + (1+r)a_0. \tag{2.23}$$

The new household's maximization problem is

$$\begin{aligned} \max_{\{c_t\}} \quad & \sum_{t=0}^{\infty} \beta^t u(c_t) \\ \text{s.t.} \quad & \sum_{t=0}^{\infty} \frac{c_t}{(1+r)^t} = \sum_{t=0}^{\infty} \frac{y_t}{(1+r)^t} + (1+r)a_0. \end{aligned} \tag{2.24}$$

The Lagrangian is

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t u(c_t) - \lambda \left[\sum_{t=0}^{\infty} \frac{c_t}{(1+r)^t} - \sum_{t=0}^{\infty} \frac{y_t}{(1+r)^t} - (1+r)a_0 \right]. \tag{2.25}$$

The first-order conditions for two subsequent periods are

$$\begin{aligned} \beta^t u'(c_t) - \lambda \frac{1}{(1+r)^t} &= 0 \\ \beta^{t+1} u'(c_{t+1}) - \lambda \frac{1}{(1+r)^{t+1}} &= 0. \end{aligned} \tag{2.26}$$

Solving each for λ gives $\lambda = \beta^t(1+r)^t u'(c_t) = \beta^{t+1}(1+r)^{t+1}u'(c_{t+1})$. Dividing by $\beta^t(1+r)^t$,

$$u'(c_t) = \beta(1+r)\, u'(c_{t+1}). \tag{2.27}$$

(2.27) is the *Euler equation*, and the interpretation is the same as the case of two periods; that is, this informs the trade-off that the households face between changing consumption in the period t and consumption in the period $t+1$.

3 The Overlapping-Generations Model

This model is based on Chapter 6 from Wickens' book [20]. Now, there is no longer a representative-agent model, which has all firms and households as the same; instead, there will be two different kinds of people, the old and the young. Every person lives for two periods. This way, the young have to decide for two periods, while the old decide only for one period. The young work, while the old retire. So, the population at the period t for the Overlapping-Generations Model (OLG) is:

$$N_t = N_{1,t} + N_{2,t}, \tag{3.1}$$

where $N_{1,t}$ represents the young population at period t, while $N_{2,t}$ indicates the old population. So, in the period t, the old generation is the young generation at period $t-1$. Therefore, $N_t = N_{1,t} + N_{1,t-1}$. Consider the growth of the population at the fixed rate n. Thus, $N_{1,t} = (1+n)N_{1,t-1}$.

$$N_t = N_{1,t} + \frac{N_{1,t}}{1+n} = \frac{2+n}{1+n}\, N_{1,t}, \qquad \text{so} \qquad \frac{N_{1,t}}{N_t} = \frac{1+n}{2+n}. \tag{3.2}$$

Since the consumption of a young and of an old person at time t are, respectively, $c_{1,t}$ and $c_{2,t}$, aggregate consumption of each cohort is:

$$C_{i,t} = c_{i,t} N_{i,t}, \qquad i = 1, 2. \tag{3.3}$$

Total consumption is then[2]:

$$\begin{aligned} C_t &= C_{1,t} + C_{2,t} \\ &= \left(c_{1,t} + \frac{1}{1+n} c_{2,t} \right) N_{1,t}. \end{aligned} \tag{3.4}$$

The resource constraint at time t is:

$$Y_t = \left(c_{1,t} + \frac{1}{1+n} c_{2,t} \right) N_{1,t} + K_{t+1} - (1-\delta) K_t. \tag{3.5}$$

Since only the young supply labor, it is convenient to normalize by the size of the working generation $N_{1,t}$ rather than by total population. Define $k_t = K_t / N_{1,t}$ and $y_t = Y_t / N_{1,t}$. Dividing (3.5) by $N_{1,t}$ and using $N_{1,t+1} = (1+n)N_{1,t}$:

$$\begin{aligned} y_t &= c_{1,t} + \frac{1}{1+n} c_{2,t} + \frac{K_{t+1}}{N_{1,t}} - (1-\delta)\frac{K_t}{N_{1,t}} \\ &= c_{1,t} + \frac{1}{1+n} c_{2,t} + \frac{N_{1,t+1}}{N_{1,t}} \frac{K_{t+1}}{N_{1,t+1}} - (1-\delta) k_t \\ &= c_{1,t} + \frac{1}{1+n} c_{2,t} + (1+n) k_{t+1} - (1-\delta) k_t. \end{aligned}$$

The production function is

$$Y_t = F(K_t, N_{1,t}), \tag{3.6}$$

assumed to have constant returns to scale, so that

$$y_t = \frac{Y_t}{N_{1,t}} = F\left(\frac{K_t}{N_{1,t}}, 1 \right) = f(k_t). \tag{3.7}$$

[2]Recall that $N_{2,t} = N_{1,t-1} = N_{1,t}/(1+n)$ is the old population at period t.

Hence the resource constraint per worker is

$$f(k_t) = c_{1,t} + \frac{1}{1+n}c_{2,t} + (1+n)k_{t+1} - (1-\delta)k_t. \tag{3.8}$$

Quantities per capita follow from (3.2): for any aggregate X_t, $X_t/N_t = \frac{1+n}{2+n}(X_t/N_{1,t})$. In particular, output per capita is $\frac{1+n}{2+n}f(k_t)$.

The rate of return to capital, for the maximization of profit, is equal to the net marginal product of the capital:

$$r_t = f'(k_t) - \delta \tag{3.9}$$

Consider constant returns to scale; thus, the wage for the young, the only ones who work in the economy, is:

$$w_t = f(k_t) - k_t f'(k_t) \tag{3.10}$$

Since the young do not work at $t+1$, they need savings at t to consume at $t+1$, so:

$$s_t = w_t - c_{1,t} \tag{3.11}$$

At time $t+1$, when they will be old, all income will be consumed, and this income will be equal to $(1+r_{t+1})s_t$. Thus, the intertemporal budget constraint is expressed as:

$$c_{1,t} + \frac{c_{2,t+1}}{1+r_{t+1}} = w_t. \tag{3.12}$$

We know that:

$$\begin{aligned} \Delta K_{t+1} &= Y_t - c_{1,t}N_{1,t} - c_{2,t}N_{1,t-1} - \delta K_t \\ &= w_t N_{1,t} + r_t K_t - c_{1,t}N_{1,t} - c_{2,t}N_{1,t-1}, \end{aligned} \tag{3.13}$$

since $c_{1,t} = w_t - s_t$ and $c_{2,t} = (1+r_t)s_{t-1}$:

$$\begin{aligned} K_{t+1} - K_t &= w_t N_{1,t} + r_t K_t - (w_t - s_t)N_{1,t} - ((1+r_t)s_{t-1})N_{1,t-1} \\ K_{t+1} - K_t &= w_t N_{1,t} + r_t K_t - w_t N_{1,t} + s_t N_{1,t} - (1+r_t)s_{t-1}N_{1,t-1} \\ K_{t+1} - K_t &= r_t K_t + s_t N_{1,t} - (1+r_t)s_{t-1}N_{1,t-1} \\ K_{t+1} &= K_t + r_t K_t + s_t N_{1,t} - (1+r_t)s_{t-1}N_{1,t-1} \\ K_{t+1} - s_t N_{1,t} &= K_t(1+r_t) - (1+r_t)s_{t-1}N_{1,t-1} \\ K_{t+1} - s_t N_{1,t} &= (1+r_t)[K_t - s_{t-1}N_{1,t-1}] \end{aligned}$$

Defining $x_t = K_t - s_{t-1}N_{1,t-1}$, this reads $x_{t+1} = (1+r_t)x_t$, which explodes unless $x_t = 0$. Hence the only non-explosive solution is

$$K_{t+1} = s_t N_{1,t}, \tag{3.14}$$

That is, the demand for capital must be equal to the supply of savings. Rearranging (3.14),

$$s_t = \frac{K_{t+1}}{N_{1,t}} = (1+n)k_{t+1} \tag{3.15}$$

Plugging (3.15) into $c_{2,t+1} = (1+r_{t+1})(w_t - c_{1,t})$,

$$c_{2,t+1} = (1 + r_{t+1})(1 + n)k_{t+1} \tag{3.16}$$

Note that (3.16) indicates that the consumption for the old is related to the next period capital stock.

Consider that the preferences of both generations are the same and equal:

$$\mathcal{U} = u(c_{1,t}) + \beta u(c_{2,t+1}) \tag{3.17}$$

This way, the maximization problem is described by:

$$\begin{aligned} \max_{c_{1,t},\, c_{2,t+1}} \quad & \mathcal{U} = u(c_{1,t}) + \beta u(c_{2,t+1}) \\ \text{s.t.} \quad & c_{2,t+1} = (1 + r_{t+1})(w_t - c_{1,t}) \end{aligned} \tag{3.18}$$

This way, the Lagrangian is

$$\mathcal{L} = u(c_{1,t}) + \beta u(c_{2,t+1}) - \lambda[c_{2,t+1} - (1 + r_{t+1})(w_t - c_{1,t})] \tag{3.19}$$

The F.O.C. are:

$$\begin{aligned} \lambda &= \frac{u'(c_{1,t})}{1 + r_{t+1}} \\ \lambda &= \beta u'(c_{2,t+1}) \\ \therefore u'(c_{1,t}) &= \beta(1 + r_{t+1})u'(c_{2,t+1}) \end{aligned} \tag{3.20}$$

which is the Euler equation for the Overlapping Generations Model.

Assume that $\beta = \frac{1}{1+\theta}$. Moreover, consider $u(c_{i,t}) = \frac{c_{i,t}^{1-\sigma}-1}{1-\sigma}$. Consequently, $u'(c_{i,t}) = c_{i,t}^{-\sigma}$. Thus,

$$c_{2,t+1} = \left(\frac{1 + r_{t+1}}{1 + \theta}\right)^{\frac{1}{\sigma}} c_{1,t} \tag{3.21}$$

Note that if the $r_{t+1} > \theta$, then the interest rate is bigger than the discount rate, i.e., the interest paid more than compensates the consumer for deferring consumption to a future period; consequently, $c_{1,t} < c_{2,t+1}$. If $r_{t+1} < \theta$, the opposite holds.

3.1 Short-Run Dynamics and Long-Run Equilibrium

First of all, plugging (3.21) into (3.11), results in:

$$s_t = w_t - \left(\frac{1 + r_{t+1}}{1 + \theta}\right)^{-\frac{1}{\sigma}} c_{2,t+1}$$

From (3.11) and (3.12), $c_{2,t+1} = (1 + r_{t+1})s_t$, so

$$
\begin{aligned}
c_{2,t+1} &= (1+r_{t+1})\left[w_t - \left(\frac{1+r_{t+1}}{1+\theta}\right)^{-\frac{1}{\sigma}} c_{2,t+1}\right] \\
c_{2,t+1} &= (1+r_{t+1})(w_t) - (1+r_{t+1})\left(\frac{1+r_{t+1}}{1+\theta}\right)^{-\frac{1}{\sigma}} c_{2,t+1} \\
c_{2,t+1} &+ (1+r_{t+1})\left(\frac{1+r_{t+1}}{1+\theta}\right)^{-\frac{1}{\sigma}} c_{2,t+1} = (1+r_{t+1})(w_t) \\
c_{2,t+1} &\left(1 + \frac{(1+r_{t+1})^{1-\frac{1}{\sigma}}}{(1+\theta)^{-\frac{1}{\sigma}}}\right) = (1+r_{t+1})(w_t) \\
c_{2,t+1} &= \frac{(1+r_{t+1})(w_t)}{1+\frac{(1+r_{t+1})^{1-\frac{1}{\sigma}}}{(1+\theta)^{-\frac{1}{\sigma}}}}
\end{aligned}
$$

Substituting w_t from (3.10) and equating with (3.16),

$$
\begin{aligned}
\frac{(1+r_{t+1})(f(k_t) - k_t f'(k_t))}{1+\frac{(1+r_{t+1})^{1-\frac{1}{\sigma}}}{(1+\theta)^{-\frac{1}{\sigma}}}} &= (1+r_{t+1})(1+n)k_{t+1} \\
\frac{(f(k_t) - k_t f'(k_t))}{1+\frac{(1+r_{t+1})^{1-\frac{1}{\sigma}}}{(1+\theta)^{-\frac{1}{\sigma}}}} &= (1+n)k_{t+1} \\
k_{t+1} &= \frac{(f(k_t) - k_t f'(k_t))}{\left\{(1+n)\left[1+\frac{(1+r_{t+1})^{1-\frac{1}{\sigma}}}{(1+\theta)^{-\frac{1}{\sigma}}}\right]\right\}}
\end{aligned}
\tag{3.22}
$$

A solution for (3.22) could be found with a logarithmic function of utility, which implies $\sigma = 1$ and, consequently, eliminates r_{t+1}. Assuming a Cobb–Douglas function means $f(k_t) = k_t^{\alpha}$ and $f'(k_t) = \alpha k_t^{-(1-\alpha)}$. This way,

$$
\begin{aligned}
k_{t+1} &= \frac{1}{(1+n)(2+\theta)}[f(k_t) - k_t f'(k_t)] \\
k_{t+1} &= \phi k_t^{\alpha},
\end{aligned}
\tag{3.23}
$$

where $\phi = (1-\alpha)/[(1+n)(2+\theta)]$. The stability of (3.23) depends on $-1 < \partial k_{t+1}/\partial k_t < 1$. Using (3.23) and assuming $k_{t+1} = k_t = k^*$ it is possible to find the value of k^* in the steady state, that is,

$$
k^* = \left[\frac{1-\alpha}{(1+n)(2+\theta)}\right]^{\frac{1}{(1-\alpha)}} \tag{3.24}
$$

The convergence around the fixed point k^* is described by the following equation:

$$
\Delta k_{t+1} = -(1-\alpha)(k_t - k^*) \tag{3.25}
$$

which follows from a first-order Taylor expansion of (3.23) around the fixed point k^*. Also it is important to point out that around k^*, $\partial k_{t+1}/\partial k_t = \alpha < 1$. The following graph describes the convergence to the steady state or k^*. Before reading the graphic, it is important to consider (3.23). The straight line represents points where $k_{t+1} = k_t$, which happens in the steady state. Starting from point A, it must run vertically to the

curve $k_{t+1} = \phi k_t^{\alpha}$, which is concave, and find the value of k_{t+1}. Since the value of k_{t+1} was identified, one must run horizontally to the 45-degree line and find the value of k_t that is the same as k_{t+1}. Thus, the process is repeated until point B is achieved, then the process stops since $k_{t+1} = k_t$ and there is no space to continue the process.

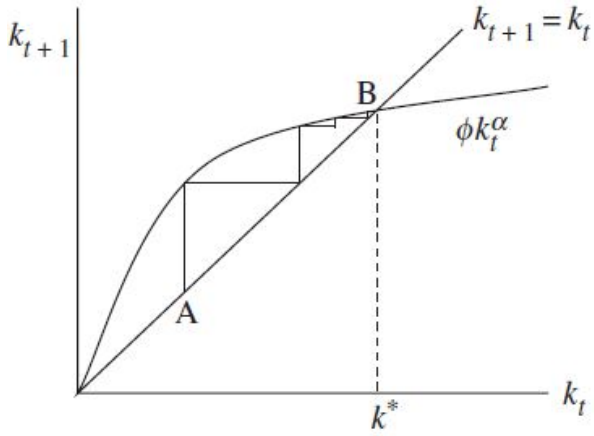


Figure 3: Source: [20]

Consider now the general CRRA case, $\sigma \neq 1$. Note that the intertemporal elasticity of substitution is $1/\sigma$, so a higher σ means a *lower* elasticity of substitution. From (3.21), $c_{2,t+1} = \big[(1 + r_{t+1})/(1 + \theta)\big]^{1/\sigma} c_{1,t}$. Define

$$Z_{t+1} := \frac{1}{1 + r_{t+1}} \left[\frac{1 + r_{t+1}}{1 + \theta}\right]^{\frac{1}{\sigma}}, \qquad \text{so that} \qquad \frac{c_{2,t+1}}{1 + r_{t+1}} = Z_{t+1}\, c_{1,t}.$$

The lifetime budget constraint (3.12) becomes $c_{1,t}(1 + Z_{t+1}) = w_t$, hence

$$s_t = w_t - c_{1,t} = w_t \frac{Z_{t+1}}{1 + Z_{t+1}}, \qquad k_{t+1} = \frac{s_t}{1 + n} = \frac{w_t}{1 + n} \frac{Z_{t+1}}{1 + Z_{t+1}},$$

using (3.15). Holding w_t and r_{t+1} fixed, k_{t+1} increases with Z_{t+1}. Taking logarithms,

$$\ln Z_{t+1} = \frac{1}{\sigma} \ln \frac{1 + r_{t+1}}{1 + \theta} - \ln(1 + r_{t+1}).$$

If $r_{t+1} > \theta$, then $\ln\big[(1 + r_{t+1})/(1 + \theta)\big] > 0$, and Z_{t+1} (hence saving and k_{t+1}) decreases when σ increases. In words: when the return on capital exceeds the discount rate, a lower elasticity of substitution (higher σ) makes the household smooth consumption more and save less, which lowers capital per worker. (If $r_{t+1} < \theta$ the effect is reversed.) Similarly, a larger population growth rate n reduces k_{t+1}, because k_{t+1} is proportional to $1/(1 + n)$. These are partial effects, for given w_t and r_{t+1}; in the steady state both prices depend on k^*.

3.2 Pensions

There are two different types of pensions that will be analyzed in these notes: the fully funded pensions, which are characterized by *"get your money, invest and give it back"*; and the Pay As You Go (PAYG) pensions, *"get your money and pay the old"*.

3.2.1 Fully Funded Pensions

In the model named as Fully Funded (FF), the pensions are provided from personal savings. There is no need to add the government in this model; however, when this occurs, the government taxes the young, invests this money, and then pays it back to the young when they become old. Assume the taxes at period t as τ_t and the investment returned for the young when they are old as p_{t+1}. This can be represented by:

$$p_{t+1} = (1 + r_{t+1})\tau_t \tag{3.26}$$

In this scheme, the budget constraint for the young generation is:

$$s_t = w_t - c_{1,t} - \tau_t \tag{3.27}$$

The consumption in the second period is:

$$\begin{aligned} c_{2,t+1} &= (1 + r_{t+1})s_t + p_{t+1}, \qquad \text{or} \\ c_{2,t+1} &= (1 + r_{t+1})(s_t + \tau_t) \end{aligned} \tag{3.28}$$

This way, the intertemporal budget constraint is:

$$c_{2,t+1} = (1 + r_{t+1})(w_t - c_{1,t}). \tag{3.29}$$

The total economy constraint at time t is:

$$\begin{aligned} &\Delta K_{t+1} = w_t N_{1,t} + r_t K_t - c_{1,t} N_{1,t} - c_{2,t} N_{1,t-1} \\ &K_{t+1} - K_t = w_t N_{1,t} + r_t K_t - (w_t - s_t - \tau_t) N_{1,t} - (1 + r_t)(s_{t-1} + \tau_{t-1}) N_{1,t-1} \\ &K_{t+1} - (s_t + \tau_t) N_{1,t} = (1 + r_t)[K_t - (s_{t-1} + \tau_{t-1}) N_{1,t-1}] \end{aligned}$$

Taking the per-worker value results in

$$s_t + \tau_t = (1 + n)k_{t+1}.$$

In (3.29) only the sum $s_t + \tau_t$ matters for the household, so the household effectively chooses total saving. If the desired total saving $(1 + n)k_{t+1}$ is at least τ_t, private saving falls one for one with the tax, $s_t = (1 + n)k_{t+1} - \tau_t \geq 0$, and capital accumulation is the same as without pensions. Only if the tax exceeds the desired saving, $\tau_t > (1 + n)k_{t+1}$, does the constraint $s_t \geq 0$ bind: the pension then forces the household to save more than it wishes, and capital accumulation increases.

3.2.2 Unfunded Pensions (PAYG)

In the pay-as-you-go (PAYG) scheme the government taxes the young and immediately transfers the proceeds to the old of the same period. Let τ_t be the tax paid by each young person and p_t the pension received by each old person at t. Budget balance in every period requires

$$\tau_t N_{1,t} = p_t N_{2,t} \quad \Longrightarrow \quad p_t = \frac{N_{1,t}}{N_{2,t}}\,\tau_t = (1 + n)\,\tau_t, \tag{3.30}$$

since $N_{2,t} = N_{1,t-1} = N_{1,t}/(1 + n)$. The contribution of a young person is not invested: it finances the current old, and the contributor is paid back by the next young generation, which is $(1 + n)$ times larger.

Consumption of the old at $t+1$ is private saving with interest plus the pension:

$$c_{2,t+1} = (1 + r_{t+1})\, s_t + p_{t+1} = (1 + r_{t+1})\, s_t + (1 + n)\, \tau_{t+1}. \tag{3.31}$$

Hence the contribution earns the implicit return n (it pays $(1+n)\tau_{t+1}$ for τ_t when τ is constant), while private saving earns r_{t+1}. With $s_t = w_t - c_{1,t} - \tau_t$, the intertemporal budget constraint is

$$c_{2,t+1} = (1 + r_{t+1})\,(w_t - c_{1,t} - \tau_t) + (1 + n)\, \tau_{t+1}. \tag{3.32}$$

Capital accumulation. Aggregate saving finances the change in the capital stock, so

$$K_{t+1} - K_t = w_t N_{1,t} + r_t K_t - c_{1,t} N_{1,t} - c_{2,t} N_{1,t-1}.$$

The young at t consume $c_{1,t} = w_t - s_t - \tau_t$, and the old at t (who saved at $t-1$) consume $c_{2,t} = (1+r_t)s_{t-1} + (1+n)\tau_t$. Substituting,

$$\begin{aligned} K_{t+1} - K_t &= w_t N_{1,t} + r_t K_t - (w_t - s_t - \tau_t) N_{1,t} - \big[(1+r_t)s_{t-1} + (1+n)\tau_t\big] N_{1,t-1} \\ &= r_t K_t + (s_t + \tau_t) N_{1,t} - (1 + r_t)\, s_{t-1} N_{1,t-1} - (1+n)\, \tau_t N_{1,t-1}. \end{aligned} \tag{3.33}$$

Because $(1+n)N_{1,t-1} = N_{1,t}$, the last term equals $\tau_t N_{1,t}$ and cancels the tax term:

$$\begin{aligned} K_{t+1} - K_t &= r_t K_t + s_t N_{1,t} - (1 + r_t)\, s_{t-1} N_{1,t-1} \\ K_{t+1} - s_t N_{1,t} &= (1 + r_t)\big[K_t - s_{t-1} N_{1,t-1}\big]. \end{aligned}$$

This holds for any path of τ_t. Define $x_t \coloneqq K_t - s_{t-1} N_{1,t-1}$, so that $x_{t+1} = (1+r_t)x_t$. If $r_t > 0$ the sequence explodes unless $x_t = 0$. The only non-explosive solution is therefore $K_{t+1} = s_t N_{1,t}$: capital is financed by private saving alone, since taxes are consumed, not invested. Dividing by $N_{1,t}$ and using $N_{1,t+1} = (1+n)N_{1,t}$ gives, per young person,

$$s_t = (1 + n)\, k_{t+1}. \tag{3.34}$$

Compare with the fully funded scheme, where $s_t + \tau_t = (1+n)k_{t+1}$: there the pension adds to the capital stock, here it does not.

Household problem. The young choose $c_{1,t}$ and $c_{2,t+1}$ taking w_t, r_{t+1}, τ_t and τ_{t+1} as given:

$$\mathcal{L} = u(c_{1,t}) + \beta\, u(c_{2,t+1}) - \lambda\Big[c_{2,t+1} - (1 + r_{t+1})(w_t - c_{1,t} - \tau_t) - (1+n)\tau_{t+1}\Big].$$

The first-order conditions are

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial c_{1,t}} &= u'(c_{1,t}) - \lambda(1 + r_{t+1}) = 0 \quad &\Longrightarrow \quad \lambda &= \frac{u'(c_{1,t})}{1 + r_{t+1}}, \\ \frac{\partial \mathcal{L}}{\partial c_{2,t+1}} &= \beta\, u'(c_{2,t+1}) - \lambda = 0 \quad &\Longrightarrow \quad \lambda &= \beta\, u'(c_{2,t+1}), \end{aligned}$$

and equating the two expressions for λ gives the Euler equation

$$u'(c_{1,t}) = \beta(1 + r_{t+1})\, u'(c_{2,t+1}).$$

With CRRA utility $u(c) = c^{1-\sigma}/(1-\sigma)$, so $u'(c) = c^{-\sigma}$, and $\beta = 1/(1+\theta)$,

$$c_{1,t}^{-\sigma} = \beta(1+r_{t+1})\, c_{2,t+1}^{-\sigma} \quad \Longrightarrow \quad c_{1,t} = \Big[\frac{1+r_{t+1}}{1+\theta}\Big]^{-\frac{1}{\sigma}} c_{2,t+1}.$$

From now on take $\sigma = 1$ (logarithmic utility), so $c_{1,t} = \dfrac{1+\theta}{1+r_{t+1}}\, c_{2,t+1}$, and a constant tax $\tau_t = \tau$. Substituting in (3.32):

$$\begin{aligned} c_{2,t+1} &= (1+r_{t+1})\Big(w_t - \frac{1+\theta}{1+r_{t+1}}\, c_{2,t+1} - \tau\Big) + (1+n)\tau \\ &= (1+r_{t+1})(w_t - \tau) - (1+\theta)\, c_{2,t+1} + (1+n)\tau \\ (2+\theta)\, c_{2,t+1} &= (1+r_{t+1})\, w_t + (n - r_{t+1})\, \tau, \end{aligned}$$

that is,

$$c_{2,t+1} = \frac{1+r_{t+1}}{2+\theta}\, w_t + \frac{n - r_{t+1}}{2+\theta}\, \tau. \tag{3.35}$$

Law of motion of capital. Solve (3.31) for private saving,

$$s_t = \frac{c_{2,t+1} - (1+n)\tau}{1+r_{t+1}},$$

and use (3.34), $k_{t+1} = s_t/(1+n)$:

$$k_{t+1} = \frac{c_{2,t+1}}{(1+n)(1+r_{t+1})} - \frac{\tau}{1+r_{t+1}}.$$

Insert (3.35) in the first term:

$$k_{t+1} = \frac{w_t}{(2+\theta)(1+n)} + \frac{(n - r_{t+1})\,\tau}{(2+\theta)(1+n)(1+r_{t+1})} - \frac{\tau}{1+r_{t+1}}.$$

The two terms in τ share the denominator $(2+\theta)(1+n)(1+r_{t+1})$, and in the numerator

$$(n - r_{t+1}) - (2+\theta)(1+n) = -\big[(2+\theta) + r_{t+1} + (1+\theta)\, n\big].$$

With $f(k) = k^\alpha$ we have $w_t = (1-\alpha)k_t^\alpha$ and $r_{t+1} = \alpha k_{t+1}^{\alpha-1} - \delta$, so

$$k_{t+1} = \phi\, k_t^\alpha - \xi_{t+1}\, \tau, \qquad \phi = \frac{1-\alpha}{(2+\theta)(1+n)}, \qquad \xi_{t+1} = \frac{(2+\theta) + r_{t+1} + (1+\theta)\, n}{(2+\theta)(1+n)(1+r_{t+1})} > 0. \tag{3.36}$$

(As in the case without pensions, r_{t+1} depends on k_{t+1}, so (3.36) defines k_{t+1} implicitly.)

Interpretation. Since $\xi_{t+1} > 0$, an increase in the PAYG tax always shifts the law of motion of capital downward and lowers the steady-state capital stock, whatever the sign of $n - r_{t+1}$. The reason is that contributions are consumed by the current old and not invested, while the young reduce their private saving only partially. The comparison between n and r_{t+1} matters for welfare, not for the direction of the capital effect. The term $(n - r_{t+1})\tau/(1+r_{t+1})$ is the change in lifetime resources of a young person, since the contribution earns n instead of r_{t+1}. If $r_{t+1} > n$ the pension is a bad deal for the

contributors. If $n > r_{t+1}$ (a dynamically inefficient economy, with too much capital) the lower capital stock is not a cost, and the scheme can raise welfare.

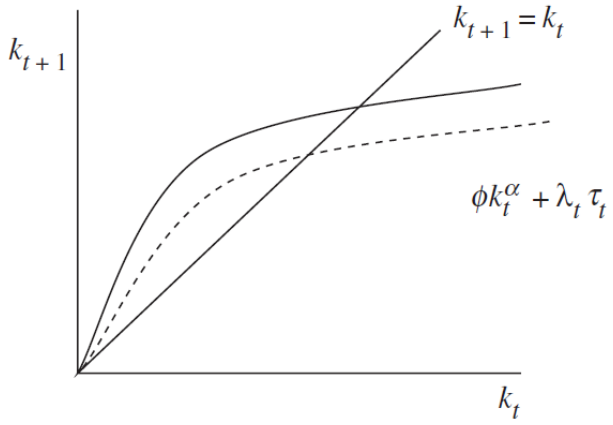


Figure 4: Source: [20]

4 Labor and Leisure

In this section, the *Labor and Leisure* model of Chapter 7 of [12] is analyzed.

4.1 Measuring the Labor Market

Some concepts hold a central place in the discussion about the development of a single country's economy; for instance, *employed* refers to those people who have worked in the past week. *Unemployed* is a person who has not worked in the last week, but has looked for a job. Finally, *out of labor force* is a person who neither has worked nor looked for a job in the last week. This way, there is:

$$\begin{aligned}
&\text{Labor force} \equiv \text{Employed} + \text{Unemployed} \\
&\text{Participation Rate} \equiv \frac{\text{Labor Force}}{\text{Population}} \\
&\text{Employment Rate} \equiv \frac{\text{Employed}}{\text{Population}} \\
&\text{Unemployment Rate} \equiv \frac{\text{Unemployed}}{\text{Labor Force}}
\end{aligned}$$

4.2 A Static Model of the Labor Market

Consider the preferences of the worker as:

$$U(c, l) = u(c) + v(l), \tag{4.1}$$

where c represents consumption and l the leisure. The functions $u(c)$ and $v(l)$ are increasing and concave; therefore, they have diminishing marginal utility. Moreover, consider L as the time spent in labor.

$$\begin{aligned}
&L + l = 1 \qquad \text{or} \\
&L = 1 - l.
\end{aligned} \tag{4.2}$$

Note that the worker must choose the best allocation of labor and leisure. Moreover, the worker has 1 unit of time to spend on labor or leisure.[3] The wage paid to the worker is w. This way, the budget restriction is,

$$c \leq w(1 - l). \tag{4.3}$$

Consequently, the problem becomes,

$$\begin{aligned}
\max_{c,l} \quad & u(c) + v(l) \\
\text{s.t.} \quad & c \leq w(1 - l)
\end{aligned} \tag{4.4}$$

To solve this problem, the Lagrange multiplier could be used; that is, the Lagrangian and the FOCs are:

[3] There is another way to report this utility function of this model, that is, $U(c, L) = u(c) - z(L)$. Note that $z(L)$ is desutility of working, and that $z(L) = -v(1 - L)$ turns this equation into (4.1).

$$
\begin{aligned}
\mathcal{L}(c, l, \lambda) &= u(c) + v(l) - \lambda[c - w(1 - l)] \\
u'(c) - \lambda &= 0 \\
v'(l) - \lambda w &= 0 \\
v'(l) &= w\, u'(c) \quad \text{or} \\
\frac{v'(l)}{u'(c)} &= w
\end{aligned}
\tag{4.5}
$$

Equation (4.5) shows that the marginal utility of one marginal unit of time allocated to leisure is $v'(l)$, while one more marginal unit of time spent at work will result in $w\,u'(c)$ utility of this consumption. Note that this is a trade-off between leisure and consumption (labor). Moreover, consumption will increase by w units, and the worker must be indifferent to allocate their last marginal unit of time between leisure and labor. The slope of the indifference curve is given by the marginal rate of substitution of consumption and leisure, $\frac{v'(l)}{u'(c)}$, and this must be equal to the w when the budget constraint curve is tangent to the indifference curve. The budget constraint curve has the slope of $-w$, i.e., one more unit of leisure costs one unit more of work. This way, as the wage is w, the consumption c will decrease by w units.[4] As shown in the following figure.

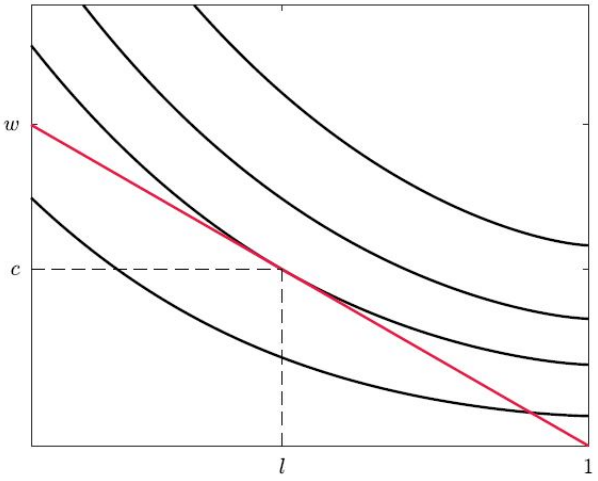


Figure 5: Source: [12]

4.2.1 The Effect of Wage Changes

Figure 5 shows that the maximum value of the x-axis is 1, as one would expect. This way, a change in the value of the wage represents a variation of the point w on this graph, and, consequently, a new indifference curve could be reached by the worker. Thus, a new allocation of c and l is chosen, and some of these new c and l are forced by the substitution effect and others by the income effect, as is depicted by the following figure:

[4]Note that the variation of c with regard to l is $-w$, that is the result of the derivative of (4.3) with respect to l.

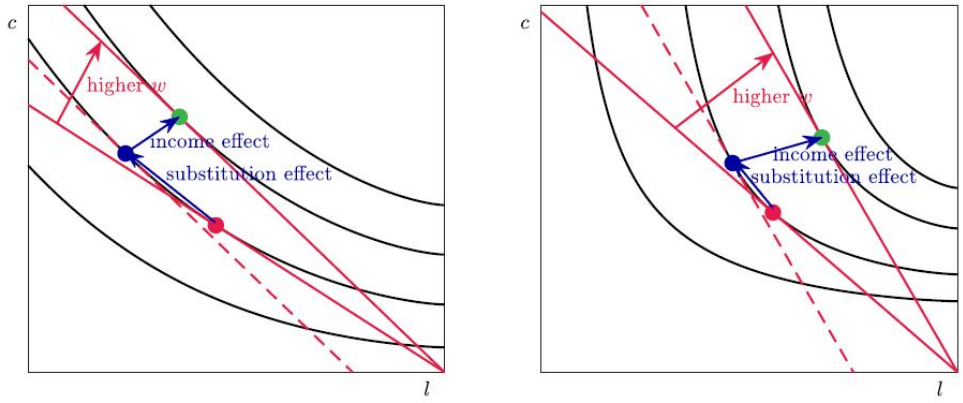


Figure 6: Source: [12]

4.2.2 Adding Taxes and Transfers

Suppose that the government collects a tax of τ on income from the worker and pays a transfer of T. The new budget constraint is:

$$c \leq w(1-l)(1-\tau) + T, \tag{4.6}$$

which demonstrates that the consumption cannot be bigger than the income plus the government transfer minus the tax deduction on the income. This new constraint could represent two types of effect. First, the transfer T indicates an upward shift in the budget constraint, which, consequently, implies that a new indifference curve will touch the budget constraint at a single point, indicating optimal worker choice. There is no change in the slope of the budget restriction; thus, the unique effect is the income effect, as the following figure indicates:

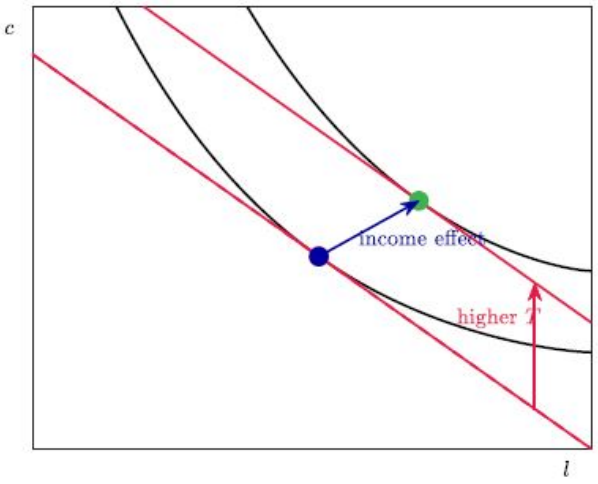


Figure 7: Source: [12]

On the other hand, the tax τ represents a change in the slope of the budget line, that is, for a given level of work, the income will be smaller, as Figure 8 shows. Note that, even if the worker chooses only leisure ($l = 1$), she will receive T, which can be used for consumption.

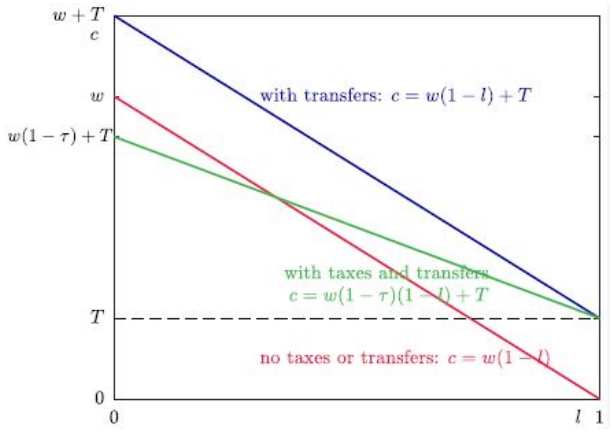


Figure 8: Source: [12]

Finally, the marginal rate of substitution of leisure for consumption is equal to the after-tax wage.

$$\frac{v'(l)}{u'(c)} = w(1-\tau) \tag{4.7}$$

4.3 A Dynamic Model

Now, suppose that there are two periods. Hence, the household must choose consumption and leisure in each period. The household maximization problem is

$$\begin{aligned} \max_{c_1,l_1,c_2,l_2} \quad & u(c_1) + v(l_1) + \beta\big[u(c_2) + v(l_2)\big] \\ \text{s.t.} \quad & c_1 + \frac{c_2}{1+r} \leq w_1(1-l_1) + \frac{w_2(1-l_2)}{1+r}. \end{aligned} \tag{4.8}$$

This means that the present value of consumption must be equal to (at most) the present value of labor income.[5] Moreover, β is the discount factor, that is, how much future consumption and leisure are valued. The Lagrangian is

$$\begin{aligned} \mathcal{L} = {} & u(c_1) + v(l_1) + \beta\big[u(c_2) + v(l_2)\big] \\ & - \lambda\left[c_1 + \frac{c_2}{1+r} - w_1(1-l_1) - \frac{w_2(1-l_2)}{1+r}\right], \end{aligned} \tag{4.9}$$

and the first-order conditions are

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial c_1} &= u'(c_1) - \lambda = 0 && \Longrightarrow \quad \lambda = u'(c_1) \\ \frac{\partial \mathcal{L}}{\partial l_1} &= v'(l_1) - \lambda w_1 = 0 && \Longrightarrow \quad \lambda = \frac{v'(l_1)}{w_1} \\ \frac{\partial \mathcal{L}}{\partial c_2} &= \beta u'(c_2) - \frac{\lambda}{1+r} = 0 && \Longrightarrow \quad \lambda = \beta(1+r)\,u'(c_2) \\ \frac{\partial \mathcal{L}}{\partial l_2} &= \beta v'(l_2) - \frac{\lambda w_2}{1+r} = 0 && \Longrightarrow \quad \lambda = \frac{\beta(1+r)\,v'(l_2)}{w_2}. \end{aligned}$$

[5] A unit of the good in the second period is worth $1/(1+r)$ units in the first period, so both sides of the constraint are expressed in present value at $t = 1$.

Equating the two expressions for λ in each period, and the expressions for λ in c_1 and c_2, one obtains

$$\frac{\upsilon'(l_t)}{u'(c_t)} = w_t \qquad \text{for } t = 1, 2 \tag{4.10}$$

$$u'(c_1) = \beta(1+r)\, u'(c_2) \tag{4.11}$$

Note that (4.10) is the same as (4.5) for each period: the household is indifferent between allocating one marginal unit of time to leisure or to work, since the income is w_t. Condition (4.11), on the other hand, is the *Euler equation* of this dynamic model of labor and leisure with two periods, that is, the distribution of consumption over time.

Assuming that υ' has an inverse function $(\upsilon')^{-1}$, since υ is strictly concave (υ' is strictly decreasing), it is possible to get the following result that helps to understand the behavior of the household if the wage rises (the last line uses (4.2)):

$$\begin{aligned} \frac{\upsilon'(l_1)}{u'(c_1)} &= w_1 \\ \upsilon'(l_1) &= w_1 u'(c_1) \\ l_1 &= (\upsilon')^{-1}[w_1 u'(c_1)] \\ L_1 &= 1 - [(\upsilon')^{-1}[w_1 u'(c_1)]] \end{aligned} \tag{4.12}$$

Compare two scenarios: a *temporary* wage rise, in which only w_1 rises, and a *permanent* one, in which both w_1 and w_2 rise. In both, $l_1 = (\upsilon')^{-1}[w_1 u'(c_1)]$ with the same w_1, so the difference comes only from c_1. In the temporary case c_1 increases only slightly, because the intertemporal mechanism smooths the extra income over the two periods. In the permanent case the household is richer in both periods and c_1 increases more. Since u' is decreasing, a larger c_1 gives a smaller $u'(c_1)$, hence a smaller $w_1 u'(c_1)$. Since υ is concave, υ' is decreasing, so $(\upsilon')^{-1}$ is decreasing, and a smaller argument gives a larger l_1. Therefore, in the permanent case l_1 is larger and $L_1 = 1 - l_1$ is smaller than in the temporary case: a permanent wage rise induces less additional labor than a temporary one, because the wealth effect (more leisure) is stronger, while the temporary rise is dominated by the substitution effect.

5 Investment

In this section, the *Investment* decision of Chapter 8 of [12] is analyzed.

First of all, the interest rate is essential for investment decisions. Consider a two-period model where a representative agent is considering building a factory, which has a total cost of 1,000 units of consumption of the unique good of the economy. The factory is built in period one and generates a revenue of 2,500 in the second period, with an operating cost of 1,400 units. The net revenue is $2{,}500 - 1{,}400 = 1{,}100$, which is greater than the cost of building the factory. Even so, the conclusion is not clear, because 1,000 units in the first period are not worth the same as 1,000 units in the second period. By building the factory, the agent gives up 1,000 units now, which could instead be lent at the interest rate r and return $1{,}000\,(1+r)$ in the second period. Thus, building the factory is worth it if $1{,}100 > 1{,}000\,(1+r)$, or, in other terms, if $r < 0.1$.[6]

5.1 Present Value

Consider an investment that pays d_t as a dividend in period t. The present value at period 0 of this stream of dividends is given by the following equation:

$$V = \sum_{t=1}^{\infty} \frac{d_t}{(1+r)^t} \tag{5.1}$$

This way, an investment project is worthwhile if the net present value is positive, that is, if $\text{NPV} \equiv V - I > 0$, where I is the cost of the investment. For the example used before,

$$\begin{aligned} \text{NPV} &= V - I \\ &= \frac{1100}{1+r} - 1000 \end{aligned}$$

That is, the investment is worthwhile only if $r < 0.1$.

Suppose that the dividends grow at a rate g forever, thus $d_{t+1} = (1+g)d_t$ or $d_t = d_1(1+g)^{t-1}$. Thus, V is:

$$\begin{aligned} V &= \sum_{t=1}^{\infty} \frac{d_1(1+g)^{t-1}}{(1+r)^t} \\ V &= \frac{d_1}{1+g} \sum_{t=1}^{\infty} \left(\frac{1+g}{1+r}\right)^t && \text{(Applying formula for a geometric sum)} \\ V &= \frac{d_1}{r-g} && \text{(simplifying, for } r > g\text{)} \end{aligned} \tag{5.2}$$

This is the Gordon growth formula: a higher interest rate lowers the present value of the dividend stream, and a higher growth rate g raises it. Sometimes it is not easy to measure the present value of, for example, rental income and compare it with the cost of the construction of a building. Returning to (5.1), there are no guarantees that the interest rate or the dividend will keep the value r and d_t in the future, since there are a lot of macroeconomic factors that will influence that. This way, analyzing asset prices

[6] A unit of the good in the first period is worth $(1+r)$ units in the second period, so a unit in the second period is worth $1/(1+r)$ units in the first.

in the stock market could be a good choice to determine if making an investment in this type of asset is a good decision, as the Tobin's Q model tries to explain:

$$Q \equiv \frac{\text{Market Value}}{\text{Book Value}} \tag{5.3}$$

where *Market Value* refers to the price of the same asset in the market, while the *Book Value* refers to the amount of the investment that will be necessary to construct a replica of this asset; therefore, if $Q > 1$, it means that investing is a good idea, on the other hand, if $Q < 1$, there must be no investment.

There is a more flexible concept about the decision of making an investment or not, that is, adjustment costs. Suppose that a firm has K capital and wants to expand its operations with the investment I. Then, the new level of capital is $K' = K + I$, i.e., the adjustment cost is the function $\Psi(I, K)$ that informs how much a firm needs to make this expansion to K', since K is its level of capital and I is the total of investment. One of the forms of this function is:

$$\Psi(I, K) = \frac{\psi I^2}{2K} \tag{5.4}$$

The derivative of this function with respect to I is $\psi \frac{I}{K}$, i.e., the marginal adjustment cost is higher when the investment I is far higher than the firm's capital.

The firm's problem for this model will be to decide how much to invest (I) today, given the capital K and its stock market value QK. Therefore:

$$\max_I Q(K + I) - \left(I + \frac{\psi I^2}{2K}\right) \tag{5.5}$$

That is, the firm wants to maximize its market value minus the cost of expansion and adjustment cost (net profit from investing), choosing I. Therefore, the first-order condition is:

$$Q - 1 - \frac{\psi I}{K} = 0$$
$$\frac{I}{K} = \frac{1}{\psi}(Q - 1)$$

That is, the investment rate relative to the size of the firm (I/K) is proportional to the extent to which Tobin's Q exceeds 1.

5.2 Risk

It is important to point out that, frequently, the dividends are not known; that is, the value of d_t depends on the state of the world at t. Suppose a two-period situation where the dividend $d(s)$ depends on the rain: $d(s) = 100$ if it rains and $d(s) = 8$ if it does not, and the probability of rain is 0.75. The interest rate is 10%. A household must decide how many units x of the asset to buy at price p in period 1. It consumes c_1 in the first period and $c_2(s)$ in the second, where s is the state and $\Pr(s)$ its probability. We look for the price p that makes the household indifferent about buying one more unit. The household's problem is

$$\max_x \; u(c_1 - xp) + \beta \sum_s \Pr(s)\, u\big(c_2(s) + x\, d(s)\big). \tag{5.6}$$

Buying x units reduces first-period consumption by xp and increases second-period consumption by $x\,d(s)$ in each state s. The first-order condition is

$$-p\,u'(c_1 - xp) + \beta \sum_s \Pr(s)\,u'\big(c_2(s) + x\,d(s)\big)\,d(s) = 0.$$

The price that makes the household indifferent about one more unit is the one for which the optimal choice is $x = 0$. Setting $x = 0$ and solving for p,

$$p = \frac{\beta \sum_s \Pr(s)\,u'(c_2(s))\,d(s)}{u'(c_1)}.$$

If the household also optimizes intertemporally, the *Euler equation* holds:

$$u'(c_1) = \beta(1+r) \sum_s \Pr(s)\,u'(c_2(s)). \tag{5.7}$$

Replacing $u'(c_1)$ in the expression for p,

$$p = \frac{1}{1+r}\,\frac{\sum_s \Pr(s)\,u'(c_2(s))\,d(s)}{\sum_s \Pr(s)\,u'(c_2(s))}. \tag{5.8}$$

With risk, the price of the asset is thus a discounted $(1/(1+r))$ weighted average of the dividends, with weights proportional to the marginal utility $u'(c_2(s))$. Since the marginal utility is decreasing, it is high when $c_2(s)$ is low, so dividends in bad states receive more weight, and assets that pay off in bad times are more valuable.

Case 1: marginal utility constant across states. If $u'(c_2(s)) = u'(c_2)$ for all s (for instance, risk-neutral household), it comes out of both sums, and since $\sum_s \Pr(s) = 1$,

$$\begin{aligned} p &= \frac{1}{1+r}\,\frac{u'(c_2)\sum_s \Pr(s)\,d(s)}{u'(c_2)\sum_s \Pr(s)} \\ &= \frac{\sum_s \Pr(s)\,d(s)}{1+r} = \frac{\mathbb{E}(d)}{1+r}. \end{aligned} \tag{5.9}$$

In the numerical example, $\mathbb{E}(d) = 0.75 \times 100 + 0.25 \times 8 = 77$, so $p = 77/1.1 = 70$.

Case 2: general case. Write the sums in (5.8) as expectations, $p = \dfrac{1}{1+r}\,\dfrac{\mathbb{E}[u'(c_2)\,d]}{\mathbb{E}[u'(c_2)]}$, where c_2 is the random second-period consumption. Using $\mathrm{Cov}(X,Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$ with $X = d$ and $Y = u'(c_2)$,

$$\mathbb{E}[u'(c_2)\,d] = \mathbb{E}(d)\,\mathbb{E}[u'(c_2)] + \mathrm{Cov}\big(d, u'(c_2)\big).$$

Dividing by $\mathbb{E}[u'(c_2)]$,

$$p = \frac{1}{1+r}\left[\mathbb{E}(d) + \frac{\mathrm{Cov}\big(d, u'(c_2)\big)}{\mathbb{E}[u'(c_2)]}\right]. \tag{5.10}$$

The covariance is positive when the dividend is high in the states in which marginal utility is high, that is, when the asset pays more in bad times. In this case $p > \mathbb{E}(d)/(1+r)$: the asset works as insurance, and the household accepts to pay more for it.[7] In the numerical example, if the household's second-period consumption is lower when it does not rain (the state with $d = 8$), the weight on that state rises and $p < 70$.

[7] Conversely, an asset that pays more in good times has a negative covariance and a price below $\mathbb{E}(d)/(1+r)$.

6 General Equilibrium

In this section, we analyze the general equilibrium model from Chapter 9 of [12]. The model's main contribution is a clearer characterization of equilibrium between firms and households, i.e., the simultaneous clearing of supply and demand across markets. In macroeconomic terms, a real equilibrium arises when every buyer obtains the quantity they desire and every seller supplies the quantity they wish. The economy is represented by three agents: a representative household that chooses consumption, savings and leisure; a representative firm that chooses between renting capital and hiring labor; and an investment firm that determines the allocation of new investment projects.

6.1 General Equilibrium in a Two-Period Economy

Consider a two-period model. In period one, K_1 is the capital of this economy, which is owned by a representative household. The production function is $F(K, L)$ and δ is the depreciation rate in the first period. In the second period, capital fully depreciates ($\delta_2 = 1$): it is the end of the world.

This way, the household problem is

$$\begin{aligned} \max_{c_1,l_1,c_2,l_2} \quad & u(c_1) + v(l_1) + \beta[u(c_2) + v(l_2)] \\ & \text{s.t.} \\ c_1 + \frac{1}{1+r}c_2 \leq\ & w_1(1-l_1) + \frac{1}{1+r}w_2(1-l_2) + K_1(1 + r_1^K - \delta) + \Pi \end{aligned} \tag{6.1}$$

Note that there are differences between this equation and that one for the labor-leisure model for the dynamic model (4.8), that is, now there is an initial capital stock (K_1), which has gross return $(1 + r_1^K - \delta)$. The other difference is a consequence of the family being the owner of the firm, so that it can get a profit in the firm's operation Π. Therefore, the profit of the firm is in the budget constraint. Π is described by $\Pi = \Pi_1^F + \frac{\Pi_2^F}{1+r} + \Pi^I$; Π_t^F represents the firm's productive profit in period t and Π^I is the investment firm's profit.[8] This way, the representative firm seeks the profit maximization, that is, its problem is:

$$\Pi_t^F = \max_{K,L} \quad F(K, L) - w_t L - r_t^K K \tag{6.2}$$

There is another firm that makes the investments; therefore, it buys $(1 - \delta)K_1$, adds I units of investment and obtains $K_2 = I + (1 - \delta)K_1$. This firm rents K_2 units of capital at a rate r_2^K. The present value of this rental income is $\frac{r_2^K}{1+r}K_2$. Therefore, the investment firm's problem is:

$$\Pi^I = \max_{I} \quad \frac{r_2^K}{1+r}[I + (1-\delta)K_1] - [(1-\delta)K_1 + I] \tag{6.3}$$

The first term of (6.3) represents the present value of rental K_2 while the second one is the cost of K_2.

6.1.1 Equilibrium Definition

For an initial capital of K_1, there will be an allocation of $c_1, l_1, c_2, l_2, I, K_2, L_1, L_2$ at prices $w_1, w_2, r_1^K, r_2^K, r$ that solves the household's problem with c_1, l_1, c_2, l_2, the production

[8] It is crucial to point out that in equilibrium $\Pi = 0$.

firm's problem with L_1, L_2, K_1, K_2, given the prices. And the problem of the investment firm with I. Moreover, in each period, the markets for goods, capital and labor clear:

- Goods:

$$\begin{aligned} F(K_1, L_1) &= c_1 + I \\ F(K_2, L_2) &= c_2 \end{aligned} \tag{6.4}$$

[9]

- Capital:

$$K_2 = K_1(1 - \delta) + I \tag{6.5}$$

[10]

- Labor:

$$L_t + l_t = 1 \tag{6.6}$$

[11]

6.1.2 Describing an Equilibrium

This subsection demonstrates the equations based on the previous sections that represent the competitive equilibrium. From the leisure-labor model, it is known that:

$$\frac{v'(l_t)}{u'(c_t)} = w_t \tag{6.7}$$

$$u'(c_1) = \beta(1 + r)u'(c_2) \tag{6.8}$$

(6.7) indicates how the household trades consumption for leisure, while (6.8) informs how the household trades consumption in period one for consumption in period 2. The firm's production has the following equations to solve its maximization problem,

$$F_K(K_t, L_t) - r_t^K = 0 \tag{6.9}$$

$$F_L(K_t, L_t) - w_t = 0 \tag{6.10}$$

(6.9) and (6.10) inform that the marginal product of capital and labor is equal to their rental price.

The investment firm's first-order condition gives:

$$1 + r = r_2^K \tag{6.11}$$

Plugging (6.10) into (6.7) results in:

$$\frac{v'(l_t)}{u'(c_t)} = F_L(K_t, L_t) \tag{6.12}$$

This indicates that the marginal rate of substitution of leisure for consumption is equal to the marginal rate of transformation of time in production.

[9] There is no Investment in the second period, only consumption.

[10] The capital in the second period will be equal to the initial capital less the depreciation plus the investment that occurred in the first period.

[11] That is, all the labor hired plus the time the household dedicates to the leisure is equal to the normalized time available, that is, 1.

Plugging (6.11) and (6.9) into (6.8) results in:

$$\frac{u'(c_1)}{\beta u'(c_2)} = F_K(K_2, L_2) \tag{6.13}$$

This indicates that the marginal rate of substitution of present consumption for future consumption, that is, how the family is willing to trade present consumption for future consumption, is equal to the marginal rate of transformation of capital in production (the marginal product of capital). (6.12) and (6.13) demonstrate what the general equilibrium looks like.

6.2 The First Welfare Theorem

The First Welfare Theorem demonstrates that the allocation of a competitive market is Pareto efficient. Suppose the existence of a benevolent social planner that has to choose the best allocation of $c_1, l_1, c_2, l_2, L_1, L_2, K_2$, he/she has to solve the following problem:

$$\begin{aligned} \max_{c_1,l_1,c_2,l_2,L_1,L_2,K_2} \quad & u(c_1) + \upsilon(l_1) + \beta[u(c_2) + \upsilon(l_2)] \\ & \text{s.t.} \\ K_2 \leq (1-\delta)K_1 + & F(K_1, L_1) - c_1 \\ c_2 \leq & F(K_2, L_2) \\ L_t \leq & 1 - l_t \qquad \text{for } t = 1, 2 \\ & K_1 \text{ given} \end{aligned} \tag{6.14}$$

Since the constraints bind at the optimum, substitute $c_1 = (1-\delta)K_1 + F(K_1, L_1) - K_2$, $c_2 = F(K_2, L_2)$ and $l_t = 1 - L_t$, this way, (6.14) becomes:

$$\max_{L_1,L_2,K_2} \quad u((1-\delta)K_1 + F(K_1, L_1) - K_2) + \upsilon(1 - L_1) + \beta[u(F(K_2, L_2)) + \upsilon(1 - L_2)] \tag{6.15}$$

So, the FOCs for L_1, L_2, K_2 are:

$$\begin{aligned} u'((1-\delta)K_1 + F(K_1, L_1) - K_2)F_L(K_1, L_1) - \upsilon'(1 - L_1) &= 0 \\ \beta[u'(F(K_2, L_2))F_L(K_2, L_2) - \upsilon'(1 - L_2)] &= 0 \\ -u'((1-\delta)K_1 + F(K_1, L_1) - K_2) + \beta u'(F(K_2, L_2))F_K(K_2, L_2) &= 0 \end{aligned}$$

It is possible to simplify these FOC to those for the model of the competitive equilibrium. Thus, it is possible to conclude that the social planner's allocation is equal to that of the competitive market, which is the optimal allocation. The reason for the equality between the social planner's allocation and the competitive market is that the former depends on the technology available. At the same time, the latter is based on the price, which reflects the technology again; thus, the result will be the same for both. This way, the First Welfare Theorem (FWT) holds when there are many goods to be consumed, technological progress, uncertainty, and different household preferences[12]. Finally, it does not hold when an economy has a monopoly or externalities.[13]

[12]For different household preferences, some adaptation must be imposed for the FWT.

[13]For more details, see [12].

6.3 General Equilibrium in an Infinite-Period Economy

For an infinite-period economy, there will be minor changes in the problem, but the steps to find the optimal allocation will be the same: describe the household's problem, the firm's problem, the investment decision, and the equilibrium. This way, the household's problem is:

$$\begin{aligned} \max_{c_t, l_t, a_{t+1}} \quad & \sum_{t=0}^{\infty} \beta^t [u(c_t) + v(l_t)] \\ & \text{s.t.} \\ a_{t+1} &= (1 + r_t)a_t + w_t(1 - l_t) + \Pi_t^F + \Pi_t^I - c_t \\ a_0 &= K_0(1+r_0^K - \delta) \qquad \text{given} \\ \lim_{T \to \infty} \frac{a_T}{(1+r)^T} &\geq 0 \qquad \text{A no-Ponzi-game condition} \end{aligned} \tag{6.16}$$

Note that $a_{t+1} = (1 + r_t)a_t + w_t(1 - l_t) + \Pi_t^F + \Pi_t^I - c_t$ is the budget constraint to the household problem. Moreover, the no-Ponzi game indicates that there are no infinite debts. Still, there is a connection between the two following periods with this budget constraint. The productive firm's problem is the same for the two-period model. On the other hand, the investment firm has the following problem:

$$\Pi_t^I = \max_I \quad \frac{r_{t+1}^K + 1 - \delta}{1 + r_{t+1}} [K_t(1 - \delta) + I] - [K_t(1 - \delta) + I] \tag{6.17}$$

The solution for those problems will be the sequences $\{c_t, l_t, I_t, K_{t+1}, L_t\}_{t=0}^{\infty}$ based on the prices $\{w_t, r_t^K, r_t\}_{t=0}^{\infty}$. Note that $\{c_t, l_t\}_{t=0}^{\infty}$, L_t, K_t, I_t solve the problems of the household, productive firm, and investment firm, respectively. Moreover, at the market equilibrium,

- Goods Market: $F(K_t, L_t) = c_t + I_t$;
- Capital Market: $K_{t+1} = K_t(1 - \delta) + I_t$;
- Labor Market: $L_t + l_t = 1$

Consequently, the first-order condition to the household's problem is given by:

$$\frac{v'(l_t)}{u'(c_t)} = w_t \tag{6.18}$$

$$u'(c_t) = \beta(1 + r_{t+1})u'(c_{t+1}) \tag{6.19}$$

The firm's problem FOC is:

$$F_K(K_t, L_t) - r_t^K = 0 \tag{6.20}$$

$$F_L(K_t, L_t) - w_t = 0 \tag{6.21}$$

The investment firm's first-order condition is:

$$r_{t+1} = r_{t+1}^K - \delta \tag{6.22}$$

Again, it is possible to plug these equations into one another to obtain some insightful results,

$$\frac{v'(l_t)}{u'(c_t)} = F_L(K_t, L_t) \tag{6.23}$$

Again, the marginal rate of substitution between leisure and consumption is equal to the marginal rate of transformation of time into production. The following equation shows that the marginal rate of substitution of consumption between the two following periods will be equal to the marginal rate of transformation of capital in production, that is,

$$\frac{u'(c_t)}{\beta u'(c_{t+1})} = (1 + F_K(K_{t+1}, L_{t+1}) - \delta) \tag{6.24}$$

6.3.1 Dynamics

Considering the long-term behavior of the economy is a common approach in economic analysis. Suppose $L_t = 1$ and $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$. This way, (6.24) becomes,

$$\frac{c_{t+1}}{c_t} = [\beta(1 + F_K(K_{t+1}, 1) - \delta)]^{\frac{1}{\sigma}} \tag{6.25}$$

This demonstrates the relation between the rate of growth of consumption between t and $t+1$ and the level of capital stock at $t+1$. Suppose that the capital stock at $t+1$ is low; thus, as a consequence of diminishing marginal product, the marginal product of capital is high. So, from (6.20) it is possible to conclude that the capital rental is also high. This way, as a consequence of (6.22), the interest rate will be high. Therefore, future consumption is more attractive than present consumption, as demonstrated by the Euler Equation, which means a growth in consumption as was supposed before.

From the assumptions about L_t and $u(c)$, the market clearing for capital between two periods is:

$$K_{t+1} = (1 - \delta)K_t + F(K_t, 1) - c_t \tag{6.26}$$

Note that (6.26) shows the relation between future capital stock, current capital stock and current consumption, when the family has a higher level of consumption at time t this means less capital stock at time $t + 1$.

The following graphic shows the dynamics of this Neoclassical Growth Model:

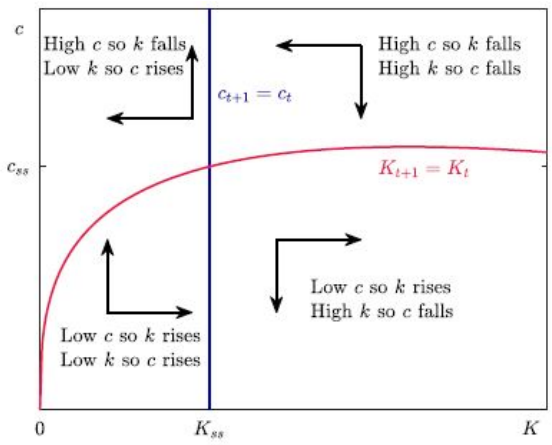


Figure 9: Source: [12]

Note that the constant consumption requires that $\frac{c_{t+1}}{c_t} = 1$. Consequently, by (6.25), $[\beta(1+F_K(K_{t+1},1)-\delta)]^{\frac{1}{\sigma}} = 1$. Therefore, $F_K(K_{t+1},1)-\delta = \frac{1}{\beta}-1$, that is, $\frac{1}{\beta}-1$ must be the interest rate to keep a constant consumption. For capital to be constant, $K_{t+1} = K_t$, i.e., $(1-\delta)K_t + F(K_t,1) - c_t = K_t$ and $c_t = F(K_t,1) - \delta K_t$, which must be the consumption level to keep the capital stock constant; if the consumption rises so that the investment will fall and the capital stock will diminish.

The path to the steady-state depends on the level of consumption at $t = 0$. K_0 is given, so there is a necessity to know c_0; the level of the initial consumption c_0, given K_0, must be in accordance with the Euler equation, otherwise the path either violates the transversality condition or exhausts the capital stock. The following graph demonstrates that:

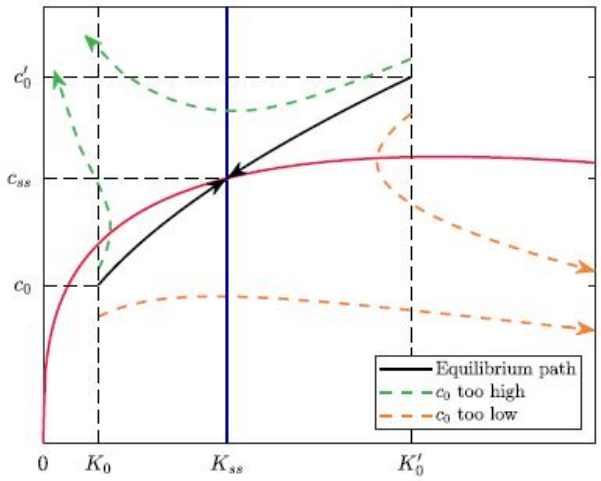


Figure 10: Source: [12]

Consider two different points c_0 for the first period consumption. The black arrows show the saddle paths, that is, the paths that bring the economy to the steady-state; the saddle path is unique. Note that, for an initial consumption higher than c_0, the green arrows indicate that all in the economy will be consumed, since the consumption is bigger than the necessary to make up the depreciation of the capital. On the other hand, if the initial consumption is lower than c_0, then all in the economy, in the long run, will be invested and nothing will be consumed, as is shown by the orange arrows. The same behavior is seen for c_0'. Thus, the saddle path will bring the economy to the saddle point, which is the steady-state, where the consumption is constant, as well as capital.

6.3.2 The Golden Rule

The Golden Rule comes from Solow's model and gives the capital stock K_{gr} that maximizes steady-state consumption. In a steady state $K_{t+1} = K_t$, so consumption is $c = F(K,1) - \delta K$. Setting its derivative with respect to K equal to zero,

$$F_K(K_{gr},1) = \delta. \tag{6.27}$$

How is the capital stock K_{ss} of the model with endogenous savings related to K_{gr}? In the steady state $c_{t+1} = c_t$, so (6.25) requires $\beta(1 + F_K(K_{ss},1) - \delta) = 1$, that is,

$$F_K(K_{ss},1) - \delta = \frac{1}{\beta} - 1. \tag{6.28}$$

Adding δ to both sides and comparing with (6.27):

$$\begin{aligned} F_K(K_{ss}, 1) &= \delta + \frac{1}{\beta} - 1 \\ &> \delta \qquad \text{as long as } \beta < 1 \\ &= F_K(K_{gr}, 1) \\ \Rightarrow \quad K_{ss} &< K_{gr}. \end{aligned} \tag{6.29}$$

The last implication comes from diminishing marginal product: F_K is decreasing in K, so if the marginal product at K_{ss} is bigger than at K_{gr}, then $K_{ss} < K_{gr}$. This result is a consequence of the household's impatience: the household prefers to consume a little more in the present, even at the expense of lower consumption in the future, which could be achieved with the Golden Rule level of capital. Note that a patient household has $\beta = 1$, so, from (6.28), $F_K(K_{ss}, 1) = \delta$, $K_{ss} = K_{gr}$ and steady-state consumption is maximized.

6.3.3 Anticipation Effects

Thinking in terms of general equilibrium helps us understand how expectations about future events affect present decisions.

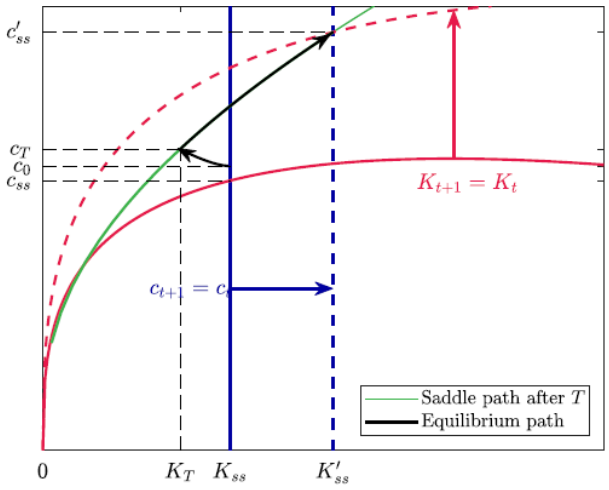


Figure 11: Source: [12]

As shown in the previous figure, if agents anticipate a future technological improvement that increases productivity, this shifts the economy's future steady state. Even before the innovation occurs, consumption rises immediately due to a wealth effect, while investment falls, causing the capital stock to shrink. As capital declines, the marginal product of capital and interest rates rise, which supports a path of gradually increasing consumption. By the time the new technology is implemented, the economy's capital and consumption levels have adjusted so that it lies exactly on the new saddle path leading to the higher steady state.

7 Calculus of Variations

The **calculus of variations** is a branch of mathematics concerned with extremizing (maximizing or minimizing) *functionals*, that is, mappings that take functions as arguments and return real numbers. In dynamic economics, this tool is fundamental for intertemporal optimization problems, such as growth models and consumption-savings decisions over time. The standard references for this section are Sydsæter *et al.* (ch. 8) [18] and Chiang (chs. 2–6) [7].

7.1 Statement of the problem

The fundamental problem of the calculus of variations consists in finding a function $y(x)$ that extremizes the functional

$$V(y) = \int_{x_1}^{x_2} F\big(x,\, y(x),\, y'(x)\big)\, dx,$$

subject to the boundary conditions

$$y(x_1) = A, \qquad y(x_2) = B,$$

where A and B are given constants. Geometrically, we seek, among all curves joining (x_1, A) to (x_2, B), the one that renders the integral as small (or as large) as possible.

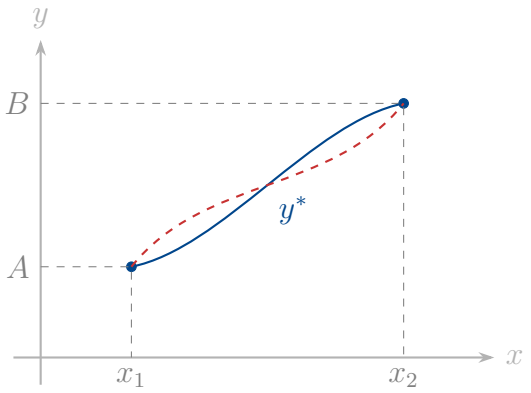


Figure 12: Among all admissible curves connecting the boundary points, we seek the curve $y^*(x)$ that extremizes the functional.

7.2 Derivation of the Euler-Lagrange equation

Suppose $y^*(x)$ is a solution of the problem and consider small variations around this curve. Write

$$y(x) = y^*(x) + \delta y(x), \qquad y'(x) = (y^*)'(x) + (\delta y)'(x),$$

where the perturbation δy vanishes at the endpoints, i.e. $\delta y(x_1) = \delta y(x_2) = 0$, since the boundary points are fixed.

The first-order variation of the functional is

$$\delta V = \int_{x_1}^{x_2} \left[\frac{\partial F}{\partial x}\, \delta x + \frac{\partial F}{\partial y}\, \delta y + \frac{\partial F}{\partial y'}\, \delta y' \right] dx.$$

Assuming $\delta x = 0$ (the independent variable is not varied), we are left with

$$\delta V = \int_{x_1}^{x_2} \frac{\partial F}{\partial y}\,\delta y\,dx + \int_{x_1}^{x_2} \frac{\partial F}{\partial y'}\,(\delta y)'\,dx.$$

Applying **integration by parts** to the second term, with $u = \partial F/\partial y'$ and $dv = (\delta y)'\,dx$, we obtain

$$\int_{x_1}^{x_2} \frac{\partial F}{\partial y'}\,(\delta y)'\,dx = \frac{\partial F}{\partial y'}\cdot\delta y\Big|_{x_1}^{x_2} - \int_{x_1}^{x_2} \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)\delta y\,dx.$$

Since δy vanishes at the endpoints, the boundary term drops out, and

$$\delta V = \int_{x_1}^{x_2} \left[\frac{\partial F}{\partial y} - \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)\right]\delta y\,dx.$$

For y^* to be an extremal, we must have $\delta V = 0$ for *every* admissible perturbation δy. By the *fundamental lemma of the calculus of variations*, this forces the integrand in brackets to vanish identically:

Euler-Lagrange equation

$$\frac{\partial F}{\partial y} - \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right) = 0.$$

Remark. Expanding the total derivative with respect to x via the chain rule,

$$\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right) = \frac{\partial^2 F}{\partial x\,\partial y'} + \frac{\partial^2 F}{\partial y\,\partial y'}\cdot y' + \frac{\partial^2 F}{\partial (y')^2}\cdot y'',$$

the Euler-Lagrange equation may be rewritten as

$$F_{y'x} + F_{y'y}\,y' + F_{y'y'}\,y'' - F_y = 0,$$

a second-order ordinary differential equation in $y(x)$.

7.3 Classical examples

Example 7.1 (Shortest distance between two points in the plane)**.** *Let $A = (x_A, y_A)$ and $B = (x_B, y_B)$ be two points in the plane. Among all curves connecting them, which has the shortest length?*

By the Pythagorean theorem, an infinitesimal arc element satisfies

$$ds^2 = dx^2 + dy^2 = dx^2\left[1 + \left(\frac{dy}{dx}\right)^2\right],$$

so that

$$ds = \sqrt{1 + (y')^2}\,dx.$$

The total length is therefore

$$S_{AB} = \int_{x_A}^{x_B} \sqrt{1 + (y')^2}\,dx,$$

with $F(x, y, y') = \sqrt{1 + (y')^2}$. Since F does not depend explicitly on y, we have $\partial F/\partial y = 0$, and the Euler-Lagrange equation reduces to

$$\frac{d}{dx}\left(\frac{y'}{\sqrt{1 + (y')^2}}\right) = 0.$$

This implies that $y'/\sqrt{1 + (y')^2}$ is constant, whence $y' = a$ is constant. Integrating,

$$y(x) = ax + b.$$

Imposing the boundary conditions $y(x_A) = y_A$ and $y(x_B) = y_B$, we obtain

$$a = \frac{y_B - y_A}{x_B - x_A}, \qquad b = y_B - \left(\frac{y_B - y_A}{x_B - x_A}\right) x_B.$$

Conclusion: the shortest curve is the straight line, as expected.

Example 7.2 (The brachistochrone problem)**.** *Posed by Johann Bernoulli in 1696, the* ***brachistochrone problem*** *asks: given two points A and B in a vertical plane, with B below and to the side of A, along which curve should a bead slide, under gravity alone, starting from rest, without friction, so as to reach B in the* shortest possible time*?*

Place A at the origin, with the x-axis horizontal and the y-axis pointing downward. Let $y(x)$ describe the curve, with $y(0) = 0$ and $y(x_B) = y_B > 0$.

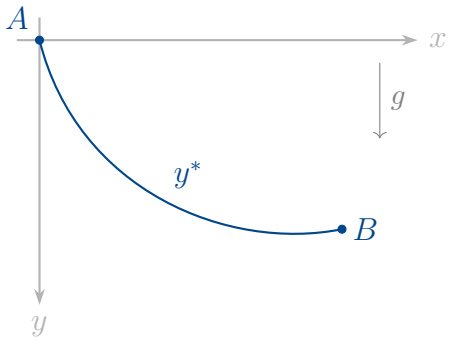


Figure 13: The brachistochrone: the curve of fastest descent between A and B under gravity.

By conservation of energy, a bead that has fallen a vertical distance y from rest has speed $v = \sqrt{2gy}$. The infinitesimal time to traverse an arc element $ds = \sqrt{1 + (y')^2}\,dx$ is

$$dt = \frac{ds}{v} = \frac{\sqrt{1 + (y')^2}}{\sqrt{2gy}}\,dx,$$

so that the total descent time is the functional

$$T(y) = \frac{1}{\sqrt{2g}} \int_0^{x_B} \sqrt{\frac{1 + (y')^2}{y}}\,dx, \qquad F(y, y') = \sqrt{\frac{1 + (y')^2}{y}}.$$

Applying the Euler-Lagrange equation, $\dfrac{\partial F}{\partial y} - \dfrac{d}{dx}\left(\dfrac{\partial F}{\partial y'}\right) = 0$, we compute

$$\frac{\partial F}{\partial y} = -\frac{1}{2} \cdot \frac{\sqrt{1 + (y')^2}}{y^{3/2}}, \qquad \frac{\partial F}{\partial y'} = \frac{y'}{\sqrt{y}\,\sqrt{1 + (y')^2}}.$$

The Euler-Lagrange equation therefore reads

$$\frac{d}{dx}\left[\frac{y'}{\sqrt{y\,(1+(y')^2)}}\right]+\frac{1}{2}\cdot\frac{\sqrt{1+(y')^2}}{y^{3/2}}=0.$$

To expand the total derivative, write $w:=1+(y')^2$*, so that* $w'=2y'y''$*. By the product and chain rules,*

$$\frac{d}{dx}\big[y'\,y^{-1/2}\,w^{-1/2}\big]=y''\,y^{-1/2}w^{-1/2}-\frac{1}{2}(y')^2\,y^{-3/2}w^{-1/2}-(y')^2y''\,y^{-1/2}w^{-3/2}.$$

Substituting into the Euler-Lagrange equation and multiplying by $y^{1/2}w^{1/2}$*,*

$$y''-\frac{(y')^2}{2y}-\frac{(y')^2\,y''}{w}+\frac{w}{2y}=0.$$

Since $1-(y')^2/w=1/w$ *and* $w-(y')^2=1$*, this reads* $\dfrac{y''}{w}+\dfrac{1}{2y}=0$*. Multiplying by* $2yw$ *gives the second-order ODE*

$$2y\,y''+(y')^2+1=0.$$

To solve it, use the standard trick $y''=y'\,\dfrac{dy'}{dy}$*, which reduces the equation to a first-order ODE in* y' *as a function of* y*:*

$$2y\,y'\,\frac{dy'}{dy}+(y')^2+1=0.$$

Letting $u:=(y')^2$*, we have* $\dfrac{du}{dy}=2y'\,\dfrac{dy'}{dy}$*, so the equation becomes linear in* u*:*

$$y\,\frac{du}{dy}+u=-1\quad\Longleftrightarrow\quad\frac{d}{dy}(y\,u)=-1.$$

Integrating, $y\,u=-y+C_1$*, hence*

$$(y')^2=\frac{C_1-y}{y}\quad\Longleftrightarrow\quad y\,(1+(y')^2)=C_1=:k.$$

This is a first-order ODE for $y(x)$*. Separating variables,*

$$dx=\sqrt{\frac{y}{k-y}}\,dy.$$

Substitute $y=\frac{k}{2}(1-\cos\theta)$*. Then* $dy=\frac{k}{2}\sin\theta\,d\theta$ *and* $k-y=\frac{k}{2}(1+\cos\theta)$*, so*

$$\sqrt{\frac{y}{k-y}}=\sqrt{\frac{1-\cos\theta}{1+\cos\theta}}=\tan\frac{\theta}{2},$$

and, using $\sin\theta=2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$*,*

$$\begin{aligned}dx&=\tan\frac{\theta}{2}\cdot\frac{k}{2}\,2\sin\frac{\theta}{2}\cos\frac{\theta}{2}\,d\theta\\&=k\sin^2\frac{\theta}{2}\,d\theta=\frac{k}{2}(1-\cos\theta)\,d\theta.\end{aligned}$$

Integrating with $x = 0$ at $\theta = 0$, we obtain the parametric solution

$$\boxed{x(\theta) = \tfrac{k}{2}(\theta - \sin\theta), \quad y(\theta) = \tfrac{k}{2}(1 - \cos\theta)}$$

which describes a ***cycloid****, the curve traced by a point on the rim of a wheel of radius $k/2$ rolling along the x-axis. The constant k is determined by the condition that the curve pass through B.*

Example 7.3 (The cake-eating problem)**.** *A consumer has an initial stock of cake $W_0 > 0$ that will be entirely consumed over a fixed horizon $[0, T]$. Let $c(t) \geq 0$ denote the consumption rate and $W(t)$ the remaining stock. The cake evolves according to*

$$\dot{W}(t) = -c(t), \qquad W(0) = W_0, \qquad W(T) = 0.$$

The consumer derives instantaneous utility $U(c)$ with $U'(c) > 0$ and $U''(c) < 0$, and discounts future utility at rate $\rho > 0$. The problem is

$$\max_{c(\cdot)} \int_0^T U(c(t))\, e^{-\rho t}\, dt.$$

Since $c = -\dot{W}$, we may rewrite the functional in terms of W alone:

$$\max_{W(\cdot)} \int_0^T \underbrace{U(-\dot{W})\, e^{-\rho t}}_{= F(t, W, \dot{W})}\, dt, \qquad W(0) = W_0, \;\; W(T) = 0.$$

Note that F does not depend explicitly on W, so $\partial F/\partial W = 0$. The Euler-Lagrange equation becomes

$$-\frac{d}{dt}\left(\frac{\partial F}{\partial \dot{W}}\right) = 0 \quad \Longleftrightarrow \quad \frac{\partial F}{\partial \dot{W}} = \text{constant}.$$

Computing the partial derivative:

$$\frac{\partial F}{\partial \dot{W}} = U'(-\dot{W})\, e^{-\rho t} \cdot (-1) = -U'(c)\, e^{-\rho t}.$$

Setting this equal to a constant and differentiating with respect to t gives

$$\frac{d}{dt}\big[U'(c)\, e^{-\rho t}\big] = 0 \quad \Longleftrightarrow \quad U''(c)\, \dot{c}\, e^{-\rho t} - \rho\, U'(c)\, e^{-\rho t} = 0,$$

which yields the ***Euler equation for consumption****:*

$$\boxed{\frac{\dot{c}}{c} = \frac{U'(c)}{c\, U''(c)} \cdot \rho = -\sigma(c) \cdot \rho,}$$

where $\sigma(c) \coloneqq -U'(c)/[c\, U''(c)] > 0$ is the intertemporal elasticity of substitution. Since $\sigma(c) > 0$ and $\rho > 0$, we conclude that $\dot{c} < 0$: the consumer optimally eats more *cake today and* less *tomorrow, at a rate dictated by the discount rate and the curvature of U.*

Special case (CRRA utility): *if $U(c) = c^{1-\theta}/(1-\theta)$ with $\theta > 0$, then $\sigma(c) = 1/\theta$ is constant, and the Euler equation integrates to*

$$c(t) = c(0)\, e^{-\rho t/\theta}.$$

Imposing the resource constraint $\int_0^T c(t)\, dt = W_0$ determines the initial consumption:

$$c(0) = \frac{W_0\, \rho/\theta}{1 - e^{-\rho T/\theta}}.$$

Example 7.4 (Ramsey-Cass-Koopmans model)**.** *Consider an economy in which the capital stock $K(t)$, consumption $c(t)$, and output $Y(t)$ evolve over time. Technology is given by $Y = f(K)$, with $f'(K) > 0$ and $f''(K) \le 0$. The resource constraint is $f(K) = c + \dot{K}$, with $K(0) = K_0$ and $K(T) = K_T$ given. Society values consumption through a utility $U(c)$ with $U'(c) > 0$ and $U''(c) < 0$. The problem is*

$$\max_{K(t)} \int_0^T U(c(t))\, e^{-\rho t}\, dt = \int_0^T \underbrace{U\big(f(K) - \dot{K}\big) e^{-\rho t}}_{=L}\, dt,$$

where $\rho > 0$ is the discount rate. Applying Euler-Lagrange with state K and velocity $\dot{K}$, $\dfrac{\partial L}{\partial K} - \dfrac{d}{dt}\left(\dfrac{\partial L}{\partial \dot{K}}\right) = 0$, we compute

$$\frac{\partial L}{\partial K} = e^{-\rho t}\, U'(c)\, f'(K), \qquad \frac{\partial L}{\partial \dot{K}} = -e^{-\rho t}\, U'(c),$$

and, using $\dot{c} = f'(K)\dot{K} - \ddot{K}$,

$$\frac{d}{dt}\big(-e^{-\rho t}\, U'(c)\big) = e^{-\rho t}\big[\rho\, U'(c) - U''(c)\, \dot{c}\big].$$

Substituting and simplifying yields

$$U'(c)\big[f'(K) - \rho\big] + U''(c)\big[f'(K)\dot{K} - \ddot{K}\big] = 0,$$

that is, $U''(c)\,\dot{c} = -U'(c)\big[f'(K) - \rho\big]$. Dividing by $c\,U''(c)$ gives the ***Keynes-Ramsey rule***

$$\frac{\dot{c}}{c} = -\frac{U'(c)}{c\,U''(c)}\big[f'(K) - \rho\big].$$

In particular, $\dot{c}/c > 0 \iff f'(K) > \rho$. Defining the intertemporal elasticity of substitution *$\sigma(c) \coloneqq -U'(c)/[c\,U''(c)] > 0$, we obtain the compact form*

$$\boxed{\frac{\dot{c}}{c} = \sigma(c)\big[f'(K) - \rho\big].}$$

7.4 Transversality conditions

So far we have assumed that both endpoints (x_1, A) and (x_2, B) were fixed. When one endpoint is **free**, the Euler-Lagrange equation alone is not enough to pin down the solution: additional boundary conditions, called **transversality conditions**, are required.

Consider the problem

$$\max_{y} V = \int_0^T F(t, y, y')\, dt, \qquad y(0) = A,$$

with T and $y(T) = y_T$ possibly free. Let $y^*(t)$ be the extremizing trajectory. Consider neighborhoods of the form

$$y(t) = y^*(t) + \delta h(t), \qquad T = T^* + \delta T,$$

so that the variation of the terminal value is $\delta y = \delta h + y^{*\prime} \cdot \delta T$ (see the example below).

The variation of the functional is

$$\delta V = \int_0^T \left[\frac{\partial F}{\partial y} - \frac{d}{dt}\left(\frac{\partial F}{\partial y'}\right)\right] \delta h\, dt + \left[\frac{\partial F}{\partial y'}\right]_T \delta y + \left[F - y'\frac{\partial F}{\partial y'}\right]_T \delta T.$$

For $\delta V = 0$, the following must hold simultaneously:

1. **Euler-Lagrange:** $\dfrac{\partial F}{\partial y} - \dfrac{d}{dt}\left(\dfrac{\partial F}{\partial y'}\right) = 0;$

2. **Transversality conditions:**

$$\left[\frac{\partial F}{\partial y'}\right]_T \delta y = 0, \qquad \left[F - y'\frac{\partial F}{\partial y'}\right]_T \delta T = 0.$$

Example 7.5 (Decomposition of δy)**.** *Let $y(t) = y^*(t) + \delta h(t)$ and $T = T^* + \delta T$. By Taylor expansion around T^*:*

$$\begin{aligned}\delta y \coloneqq y(T) - y^*(T^*) &\approx y(T^*) + y'(T^*)(T - T^*) - y^*(T^*) \\ &= (y - y^*)(T^*) + y'(T^*)\,\delta T \\ &= \delta h(T^*) + y'(T^*)\,\delta T.\end{aligned}$$

Taking $T \to T^*$,

$$\delta y = \delta h + y' \cdot \delta T.$$

Boundary configurations

The transversality conditions take specific forms depending on the geometry of the terminal boundary:

(i) Vertical terminal line (T^* given, $y(T^*)$ free): we have $\delta T = 0$ and $\delta y \neq 0$, so that

$$\left.\frac{\partial F}{\partial y'}\right|_{t=T^*} = 0.$$

(ii) Horizontal terminal line (T free, y_T given): we have $\delta T \neq 0$ and $\delta y = 0$, so that

$$\left[F - y'\frac{\partial F}{\partial y'}\right]_{t=T} = 0.$$

(iii) Terminal curve $y_T = \phi(T)$: in this case $\delta y = \phi'(T)\,\delta T$, and combining both terms yields

$$\left[F - \big(y' - \phi'(T)\big)\frac{\partial F}{\partial y'}\right]_{t=T} = 0.$$

7.5 Infinite horizon

In many economic problems, we take $T \to \infty$. The transversality condition

$$\left[F - y'\frac{\partial F}{\partial y'}\right]_{t\to\infty} \delta T + \left[\frac{\partial F}{\partial y'}\right]_{t\to\infty} \delta y_T = 0$$

splits into two cases:

(i) If $\lim\limits_{t\to\infty} y(t) = y_\infty$ is given, then $\delta y_T = 0$ and

$$\lim_{t\to\infty}\left(F - y'\frac{\partial F}{\partial y'}\right) = 0.$$

(ii) If y_∞ is free, then additionally

$$\lim_{t\to\infty} \frac{\partial F}{\partial y'} = 0.$$

Example 7.6 (Ramsey-Cass-Koopmans with infinite horizon)**.** *For*

$$F = e^{-\rho t}\, U\big(f(K) - \dot{K}\big)$$

with $c = f(K) - \dot{K}$*, we have:*

- *Case (i):* $\lim\limits_{t\to\infty} e^{-\rho t}\big[U(c) + \dot{K}\, U'(c)\big] = 0$*;*
- *Case (ii):* $\lim\limits_{t\to\infty} e^{-\rho t}\, U'(c) = 0.$

The latter is the variational form of the transversality condition *of the optimal growth model: the discounted marginal utility of terminal consumption must tend to zero. Section 8 relates it to the more common form* $\lim_{t\to\infty} e^{-\rho t}\, U'(c)\, K(t) = 0.$

8 The Ramsey-Cass-Koopmans model

The Ramsey-Cass-Koopmans model is the *benchmark* of growth macroeconomics with microfoundations. Originally developed by Frank Ramsey in 1928, and revisited independently by David Cass and Tjalling Koopmans in the 1960s, the model describes an economy in which a representative agent chooses the optimal trajectory of consumption and capital accumulation, maximizing intertemporal welfare. The canonical reference for this section is Acemoglu (2008), ch. 8 [1].

8.1 Economic environment

Consider a closed economy, without government, with a single good that can be consumed or accumulated as capital. Time is continuous and the horizon is infinite. The economy is inhabited by a representative household of constant size (normalized to one), which derives instantaneous utility $U(c)$ from per capita consumption $c(t)$, satisfying

$$U'(c) > 0, \qquad U''(c) < 0, \qquad \lim_{c \to 0^+} U'(c) = +\infty.$$

The first two conditions reflect positive and diminishing marginal utility; the third (Inada condition) ensures that consumption is strictly positive along the optimal trajectory.

Technology is given by a production function $Y = F(K, L)$ with constant returns to scale, positive and diminishing marginal returns, and satisfying the Inada conditions:

$$F_K > 0,\ F_{KK} < 0, \qquad \lim_{K \to 0} F_K = +\infty, \quad \lim_{K \to \infty} F_K = 0.$$

Normalizing $L = 1$, we write $f(k) \coloneqq F(k, 1)$, with $f'(k) > 0$ and $f''(k) < 0$.

Capital evolves according to the resource constraint

$$\dot{k}(t) = f(k(t)) - c(t) - \delta\, k(t),$$

where $\delta \geq 0$ is the depreciation rate. For simplicity, in this course we take $\delta = 0$; the general case is analogous.

8.2 The consumer's problem

The representative household solves

$$\max_{c(\cdot)} \int_0^\infty U(c(t))\, e^{-\rho t}\, dt$$

subject to

$$\dot{k}(t) = f(k(t)) - c(t), \qquad k(0) = k_0 > 0, \qquad \lim_{t \to \infty} k(t) \geq 0,$$

where $\rho > 0$ is the intertemporal discount rate. The condition $k(t) \geq 0$ in the limit prevents the household from ending up indebted at infinity (*no-Ponzi* condition).

Remark. The condition $\rho > 0$ is necessary for the functional to converge: without discounting, the integral of $U(c) \cdot 1$ over $[0, \infty)$ typically diverges for growth trajectories. Exponential discounting is what makes the problem well-posed.

8.3 Variational formulation

We eliminate c from the constraint using $c = f(k) - \dot{k}$:

$$\max_{k(\cdot)} \int_0^\infty \underbrace{U\big(f(k) - \dot{k}\big)\, e^{-\rho t}}_{=F(t,k,\dot{k})}\, dt.$$

The problem now has k as the only choice variable, and falls into the standard form of the calculus of variations, with $F(t, k, \dot{k}) = U(f(k) - \dot{k})\, e^{-\rho t}$.

8.4 Derivation via Euler-Lagrange

We apply the Euler-Lagrange equation:

$$\frac{\partial F}{\partial k} - \frac{d}{dt}\left(\frac{\partial F}{\partial \dot{k}}\right) = 0.$$

Computing each term:

$$\frac{\partial F}{\partial k} = U'(c)\, f'(k)\, e^{-\rho t}, \qquad \frac{\partial F}{\partial \dot{k}} = -U'(c)\, e^{-\rho t}.$$

The total derivative of the second term is

$$\frac{d}{dt}\big(-U'(c)\, e^{-\rho t}\big) = -U''(c)\, \dot{c}\, e^{-\rho t} + \rho\, U'(c)\, e^{-\rho t}.$$

Substituting into the Euler-Lagrange equation:

$$U'(c)\, f'(k)\, e^{-\rho t} - \big[-U''(c)\, \dot{c}\, e^{-\rho t} + \rho\, U'(c)\, e^{-\rho t}\big] = 0.$$

Dividing by $e^{-\rho t} > 0$ and rearranging,

$$U'(c)\big[f'(k) - \rho\big] + U''(c)\, \dot{c} = 0,$$

which, isolating $\dot{c}$, yields the **consumption Euler equation**:

Euler equation (Keynes-Ramsey)

$$\frac{\dot{c}}{c} = -\frac{U'(c)}{c\, U''(c)}\big[f'(k) - \rho\big] = \sigma(c)\big[f'(k) - \rho\big],$$

where $\sigma(c) \coloneqq -U'(c)/[c\, U''(c)] > 0$ is the *intertemporal elasticity of substitution*.

Remark. The economic interpretation is direct: consumption grows over time if and only if the marginal product of capital exceeds the discount rate. When $f'(k) > \rho$, it pays to postpone consumption (save), since the return on saving outweighs impatience; when $f'(k) < \rho$, the household prefers to consume now.

8.5 Transversality condition

Since the horizon is infinite and $k(\infty)$ is free (the household chooses how much capital to leave at infinity), an additional condition is needed to close the problem. Applying the transversality condition for infinite horizon of Section 7:

$$\lim_{t\to\infty} \frac{\partial F}{\partial \dot{k}} = 0 \quad \Longleftrightarrow \quad \lim_{t\to\infty} U'(c(t))\, e^{-\rho t} = 0.$$

Along the convergent path $k(t) \to k^* > 0$, this is equivalent to the form more common in the literature, which weights the shadow price of capital by the capital stock:

$$\lim_{t\to\infty} e^{-\rho t}\, U'(c(t))\, k(t) = 0.$$

This second form is the one delivered by the maximum principle (Section 10). Economically, it says that the present value of the terminal capital stock is zero: the household does not accumulate capital that it never consumes.

8.6 Dynamic system and steady state

Combining the Euler equation with the resource constraint, we obtain the autonomous two-dimensional system

$$\begin{cases} \dot{c} = \sigma(c)\, c \left[f'(k) - \rho \right], \\ \dot{k} = f(k) - c. \end{cases}$$

The steady state (k^*, c^*) is characterized by $\dot{c} = \dot{k} = 0$:

$$\boxed{f'(k^*) = \rho, \qquad c^* = f(k^*).}$$

This is the **modified golden rule**: in steady state, the marginal product of capital equals the discount rate. Compare with the *golden rule*, the capital stock that maximizes steady-state consumption. With $\delta = 0$ (and no population growth), steady-state consumption is $c = f(k)$, which is increasing in k, so there is no finite golden rule. With depreciation $\delta > 0$, steady-state consumption is $c = f(k) - \delta k$, maximized at $f'(k_{gr}) = \delta$, while the Euler equation gives $f'(k^*) = \rho + \delta$. Since $\rho > 0$ and f' is decreasing, $k^* < k_{gr}$: impatience leads agents to accumulate less capital than the level that maximizes steady-state consumption.

8.7 Phase diagram

The system above has a unique interior equilibrium point, which is a **saddle point**. In the (k, c) plane:

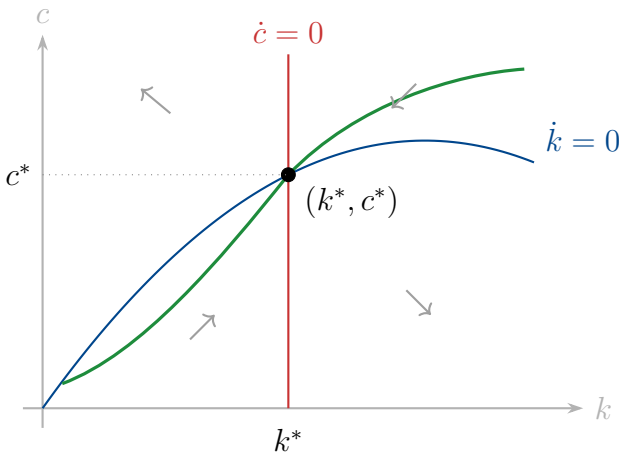


Figure 14: Phase diagram of the Ramsey-Cass-Koopmans model, drawn with $\delta > 0$, so that the curve $\dot{k} = 0$, $c = f(k) - \delta k$, has a maximum. The equilibrium (k^*, c^*) is a saddle point. The gray arrows show the direction of motion in each region; only the green trajectory (stable branch) converges to the equilibrium.

Of the infinitely many trajectories starting from $k(0) = k_0$, only one, along the **stable branch** of the saddle, converges to (k^*, c^*). The others either drive capital to zero in finite time (with c exploding) or drive consumption to zero (with k growing unboundedly). The transversality condition selects precisely the trajectory that converges to the steady state.

8.8 CRRA case

When utility is of the CRRA type (*constant relative risk aversion*),

$$U(c) = \frac{c^{1-\theta} - 1}{1 - \theta}, \qquad \theta > 0,\ \theta \neq 1,$$

(and $U(c) = \ln c$ for $\theta = 1$), the elasticity $\sigma(c) = 1/\theta$ is constant. The Euler equation simplifies to

$$\frac{\dot{c}}{c} = \frac{1}{\theta}\big[f'(k) - \rho\big].$$

This will be the form most used in subsequent sections, as it allows closed-form analytical solutions in several models.

9 Endogenous growth: the AK model

The Ramsey-Cass-Koopmans model predicts convergence to the steady state: in the long run, per capita income growth is zero (unless *exogenous* technical progress is introduced). This implication is uncomfortable, since the data show sustained per capita income growth in developed economies for over a century. The theory of **endogenous growth**, developed in the 1980s and 1990s, seeks to explain this growth from mechanisms internal to the model, without appealing to exogenous technical progress. In this section, we present the AK model, the simplest and most transparent version of this family. The main reference is Acemoglu (2008), ch. 11 [1].

9.1 The central idea

In the Ramsey model, the production function $f(k)$ satisfies the Inada conditions, in particular $\lim_{k\to\infty} f'(k) = 0$. This is what generates *diminishing returns*: as k grows, the marginal product falls, discouraging further accumulation. Eventually, $f'(k^*) = \rho$ and growth ceases.

The AK model breaks this dynamic by assuming a **linear** production function in capital:

$$f(k) = A\,k, \qquad A > 0.$$

Here, the marginal product $f'(k) = A$ is constant, regardless of how large k becomes. There are no diminishing returns, and capital accumulation can sustain growth indefinitely.

Remark. The function $f(k) = Ak$ admits several interpretations. One reading is that k includes not only physical capital but also *human capital* and other forms of reproducible capital, so that, in the aggregate, there is no fixed factor (such as raw labor) generating diminishing returns. Another reading, due to Romer (1986), is that there exist capital externalities (learning, spillovers) that offset private diminishing returns. In all interpretations, the point is that the accumulable factor has constant marginal product in the aggregate.

9.2 The problem

The consumer's problem is identical to Ramsey's, except for the production function:

$$\max_{c(\cdot)} \int_0^\infty U(c)\,e^{-\rho t}\,dt \quad \text{s.t.} \quad \dot{k} = A\,k - c, \quad k(0) = k_0, \quad \lim_{t\to\infty} k(t) \geq 0.$$

We assume CRRA utility, $U(c) = c^{1-\theta}/(1-\theta)$, with $\theta > 0$, to obtain explicit solutions.

9.3 Derivation via Euler-Lagrange

Eliminating $c = Ak - \dot{k}$, the functional to maximize is

$$\int_0^\infty \frac{(Ak - \dot{k})^{1-\theta}}{1-\theta}\,e^{-\rho t}\,dt.$$

Defining $F(t, k, \dot{k}) = \frac{(Ak-\dot{k})^{1-\theta}}{1-\theta}\,e^{-\rho t}$ and applying Euler-Lagrange,

$$\frac{\partial F}{\partial k} - \frac{d}{dt}\left(\frac{\partial F}{\partial \dot{k}}\right) = 0,$$

we obtain, via calculation identical to Ramsey's with $f'(k) = A$:

Euler equation in AK

$$\frac{\dot{c}}{c} = \frac{1}{\theta}(A - \rho) =: g.$$

The novelty is that, since A is a *constant*, the growth rate of consumption is also constant over time. There is no transition dynamics: the economy already starts on its long-run growth rate.

Remark. For the problem to be well-posed, we require $A > \rho$ (so that $g > 0$ and there is growth) and also $A - \rho < \theta A$, equivalently $\rho > (1 - \theta)A$, to ensure convergence of the functional. Under logarithmic utility ($\theta = 1$), it suffices that $\rho > 0$.

9.4 Equilibrium trajectory

Solving the Euler equation,

$$c(t) = c(0)\, e^{gt}, \qquad g = \frac{A - \rho}{\theta}.$$

Substituting into the resource constraint $\dot{k} = Ak - c$, we obtain the non-homogeneous linear ODE

$$\dot{k} - Ak = -c(0)\, e^{gt}.$$

The general solution is $k(t) = \alpha e^{At} + \beta e^{gt}$ for appropriate constants α, β. Imposing $k(0) = k_0$ and the transversality condition

$$\lim_{t\to\infty} e^{-\rho t}\, U'(c(t))\, k(t) = \lim_{t\to\infty} e^{-\rho t}\, c(t)^{-\theta}\, k(t) = 0,$$

the explosive term e^{At} is eliminated. Indeed, $c(t)^{-\theta} = c(0)^{-\theta} e^{-\theta g t}$ and $\rho + \theta g = A$, so

$$e^{-\rho t}\, c(t)^{-\theta}\, k(t) = c(0)^{-\theta}\, e^{-At}\left[\alpha e^{At} + \beta e^{gt}\right] = c(0)^{-\theta}\left[\alpha + \beta e^{-(A-g)t}\right] \xrightarrow[t\to\infty]{} c(0)^{-\theta}\, \alpha,$$

because $g < A$ (which is equivalent to $A - \rho < \theta A$, assumed above). Hence transversality requires $\alpha = 0$, and $k(0) = k_0$ gives $\beta = k_0$, resulting in

$$k(t) = k_0\, e^{gt}, \qquad c(t) = c(0)\, e^{gt},$$

with $c(0)$ determined by the initial constraint. Indeed, imposing $\dot{k}(0) = Ak_0 - c(0) = g\, k_0$, we obtain

$$c(0) = (A - g)\, k_0 = \left(A - \frac{A - \rho}{\theta}\right) k_0 = \frac{(\theta - 1)A + \rho}{\theta}\, k_0.$$

Therefore, the entire economy grows at the constant rate g:

$$\frac{\dot{c}}{c} = \frac{\dot{k}}{k} = \frac{\dot{y}}{y} = g = \frac{A - \rho}{\theta}.$$

9.5 Implications

The AK model has several implications that contrast sharply with Ramsey:

1. **Absence of transitional dynamics.** The economy starts directly on its balanced growth path; there is no convergence to a steady state because there is no steady state (in the sense of constant k).

2. **No convergence across economies.** Two economies with the same parameters (A, ρ, θ) but different initial conditions grow at the same rate g, maintaining their relative income gap forever. This contrasts with Ramsey, where economies converge to the same k^*.

3. **Growth depends on structural parameters.** Policies that affect A (total productivity), ρ (impatience), or θ (the inverse of the elasticity of intertemporal substitution) *permanently* affect the growth rate, not merely the level.

4. **Growth is endogenous.** There is no exogenous technical progress; growth emerges from the continuous accumulation of the reproducible factor k.

9.6 Limitations and extensions

The AK model is deliberately stylized. Its main limitations are:

1. The linearity $f(k) = Ak$ is a strong hypothesis with weak microeconomic justification. It is not obvious why aggregate capital would have constant marginal product;

2. Empirically, poor and rich economies do *not* grow at the same rate indefinitely; there is evidence of conditional convergence, which the pure AK model does not predict;

3. There is no role for innovation, R&D, or any microeconomic process that generates growth.

These limitations motivated extensions such as:

- **Romer (1986):** private diminishing returns, but positive externalities of aggregate capital, generating constant returns in the aggregate;

- **Lucas (1988), the Uzawa-Lucas model:** two reproducible factors, physical capital k and human capital h, with endogenous accumulation of h via time spent in education (Section 11);

- **Romer (1990):** varieties of intermediate goods created by R&D, with monopolistic competition and increasing returns derived from the non-rivalry of knowledge.

All these extensions preserve the central idea: for sustained endogenous growth to arise, some accumulable factor must have a marginal product that does not tend to zero.

10 Optimal Control

The calculus of variations handles problems in which the only function to be chosen is $y(t)$ and its derivative $y'(t)$ enters the integrand directly. Many economic problems do not have this form: there is a *state* variable $y(t)$, whose evolution is governed by a law of motion, and a *control* variable $u(t)$, chosen freely at each instant, which affects both the integrand and the law of motion. In the Ramsey and AK models we were able to eliminate the control ($c = f(k) - \dot{k}$) and reduce the problem to the variational form. This is not always possible, for instance when a control enters two laws of motion at once, as will happen in the Uzawa-Lucas model. Optimal control theory, due to Pontryagin and his collaborators in the 1950s, provides the general method. The main reference for this section is Chiang (1992), ch. 7 [8]; see also Kamien and Schwartz (1991), part II [10].

10.1 The problem

We consider

$$\max_{u(\cdot)} \; V = \int_0^T F\big(t, y(t), u(t)\big)\, dt$$

subject to

$$\dot{y}(t) = f\big(t, y(t), u(t)\big), \qquad y(0) = A, \qquad y(T) \text{ free},$$

with A and T given. The functions F and f are assumed continuously differentiable in all arguments. We write $y_T := y(T)$ for the terminal value of the state.

10.2 The Hamiltonian

Define the **Hamiltonian**

$$H(t, y, u, \lambda) := F(t, y, u) + \lambda(t)\, f(t, y, u),$$

where $\lambda(t)$ is a function to be determined, called the *costate* (or *adjoint*) variable. We will show that any solution of the problem must satisfy the following conditions.

Pontryagin's Maximum Principle (necessary conditions)

Let (y^*, u^*) solve the problem above. Then there exists a continuously differentiable function $\lambda(t)$ such that, for all $t \in [0, T]$,

(i) $u^*(t)$ maximizes $H\big(t, y^*(t), u, \lambda(t)\big)$ over all admissible u; in particular, at an interior maximum, $\dfrac{\partial H}{\partial u} = 0$;

(ii) $\dot{\lambda} = -\dfrac{\partial H}{\partial y}$ (costate equation);

(iii) $\dot{y} = \dfrac{\partial H}{\partial \lambda}$ (state equation, which is just $\dot{y} = f$);

(iv) $\lambda(T) = 0$ (transversality condition for T given and y_T free).

Remark. The name "maximum principle" refers to condition (i), which says that at each instant the optimal control *maximizes* the Hamiltonian, not merely that it is a stationary

point. The first-order condition $\partial H/\partial u = 0$ is the interior version of (i), valid when the maximum is attained in the interior of the admissible set and H is differentiable in u. When the control is constrained, for instance $u \in [0,1]$, corner solutions are possible, and (i) must be used in its full form. The derivation below, which uses small perturbations of u^*, delivers only the first-order condition; the full statement requires a different argument (the so-called needle variations), which we omit. We will meet a corner solution in the Ben-Porath model.

10.3 Derivation

The strategy is to absorb the constraint into the objective, as in static constrained optimization, and then apply the variational method of Section 7 to the resulting unconstrained functional.

Step 1: the augmented functional

Since the constraint $\dot{y} = f(t,y,u)$ holds at every t, we have, for any function $\lambda(t)$,

$$\int_0^T \lambda(t)\big[f(t,y,u) - \dot{y}\big]\,dt = 0.$$

Adding this zero to V,

$$V = \int_0^T F(t,y,u)\,dt + \int_0^T \lambda(t)\big[f(t,y,u) - \dot{y}\big]\,dt = \int_0^T \underbrace{\big[F + \lambda f\big]}_{=H}\,dt - \int_0^T \lambda\,\dot{y}\,dt.$$

Integrating the last term by parts, with λ as the function to differentiate and $\dot{y}\,dt$ as the differential,

$$-\int_0^T \lambda\,\dot{y}\,dt = -\Big[\lambda(t)\,y(t)\Big]_0^T + \int_0^T \dot{\lambda}\,y\,dt = -\lambda(T)\,y_T + \lambda(0)\,y_0 + \int_0^T \dot{\lambda}\,y\,dt.$$

Therefore

$$V = \int_0^T \Big[H(t,y,u,\lambda) + \dot{\lambda}\,y\Big]dt - \lambda(T)\,y_T + \lambda(0)\,y_0. \tag{1}$$

This expression no longer contains $\dot{y}$: the derivative of the state has been traded for the derivative of the costate.

Step 2: neighboring trajectories

Let (y^*, u^*) be the optimal pair, with optimal terminal time T^* and optimal terminal value y_T^*. We consider the one-parameter family

$$u(t) = u^*(t) + \varepsilon\,p(t), \qquad y(t) = y^*(t) + \varepsilon\,q(t), \qquad T = T^* + \varepsilon\,\Delta T, \qquad y_T = y_T^* + \varepsilon\,\Delta y_T,$$

where p, q are arbitrary continuously differentiable functions, ΔT and Δy_T are arbitrary real numbers, and $\varepsilon \in \mathbb{R}$ is small. Since $y(0) = A$ is fixed, $q(0) = 0$. We keep T and y_T as potentially free for now, so that the derivation covers all boundary cases at once; the fixed-T case is recovered by setting $\Delta T = 0$.

Substituting into (1), V becomes an ordinary function of ε:

$$V(\varepsilon) = \int_0^{T^*+\varepsilon\Delta T} \Big[H\big(t,\, y^* + \varepsilon q,\, u^* + \varepsilon p,\, \lambda\big) + \dot\lambda \,\big(y^* + \varepsilon q\big) \Big]\, dt - \lambda(T)\, y_T + \lambda(0)\, y_0,$$

where in the boundary term both $\lambda(T) = \lambda(T^* + \varepsilon\Delta T)$ and $y_T = y_T^* + \varepsilon\Delta y_T$ depend on ε. Optimality of (y^*, u^*) requires

$$\left.\frac{dV}{d\varepsilon}\right|_{\varepsilon=0} = 0.$$

Step 3: computing $dV/d\varepsilon$

We differentiate term by term. The integral has an ε-dependent upper limit and an ε-dependent integrand, so by the Leibniz rule

$$\frac{d}{d\varepsilon}\int_0^{T(\varepsilon)} G(t,\varepsilon)\, dt = G\big(T(\varepsilon),\varepsilon\big)\,\frac{dT}{d\varepsilon} + \int_0^{T(\varepsilon)} \frac{\partial G}{\partial \varepsilon}\, dt.$$

Here $G = H + \dot\lambda\, y$ and $dT/d\varepsilon = \Delta T$. The partial derivative of the integrand with respect to ε, by the chain rule, is

$$\frac{\partial G}{\partial \varepsilon} = \frac{\partial H}{\partial y}\, q + \frac{\partial H}{\partial u}\, p + \dot\lambda\, q.$$

For the boundary term, by the product rule,

$$\frac{d}{d\varepsilon}\big[-\lambda(T)\, y_T\big] = -\dot\lambda(T)\,\frac{dT}{d\varepsilon}\, y_T - \lambda(T)\,\frac{dy_T}{d\varepsilon} = -\dot\lambda(T)\,\Delta T\, y_T - \lambda(T)\,\Delta y_T.$$

The term $\lambda(0)\, y_0$ does not depend on ε. Collecting everything and evaluating at $\varepsilon = 0$ (so that $T = T^*$, $y = y^*$, $u = u^*$, $y_T = y_T^*$):

$$\left.\frac{dV}{d\varepsilon}\right|_0 = \int_0^{T^*} \left\{ \left[\frac{\partial H}{\partial y} + \dot\lambda\right] q + \frac{\partial H}{\partial u}\, p \right\} dt + \Big[H + \dot\lambda\, y\Big]_{t=T^*} \Delta T - \Big[\dot\lambda\, y\Big]_{t=T^*} \Delta T - \lambda(T^*)\,\Delta y_T.$$

The two terms in $\dot\lambda\, y$ at T^* cancel, leaving

First variation of the augmented functional

$$\left.\frac{dV}{d\varepsilon}\right|_0 = \int_0^{T^*} \left[\frac{\partial H}{\partial y} + \dot\lambda\right] q\, dt + \int_0^{T^*} \frac{\partial H}{\partial u}\, p\, dt + \Big[H\Big]_{t=T^*} \Delta T - \lambda(T^*)\,\Delta y_T = 0. \quad (2)$$

Step 4: extracting the conditions

Equation (2) must hold for every admissible variation. The variations p, ΔT and Δy_T can be chosen freely. The variation q cannot: it is the change in the state path induced by p through the law of motion. However, the function $\lambda(t)$ has not been specified so far, and we are free to choose it.

- Choose λ so that the coefficient of q vanishes,

$$\frac{\partial H}{\partial y} + \dot\lambda = 0 \quad\Longleftrightarrow\quad \dot\lambda = -\frac{\partial H}{\partial y}.$$

With this choice the first integral in (2) is zero whatever q is.

- Taking $\Delta T = 0$, $\Delta y_T = 0$ and p arbitrary, the fundamental lemma of the calculus of variations gives
$$\frac{\partial H}{\partial u} = 0.$$
- With T given, $\Delta T = 0$ and the third term is absent. If y_T is free, Δy_T is arbitrary and we need
$$\lambda(T^*) = 0.$$
- Finally, $\partial H/\partial \lambda = f(t, y, u)$ by the definition of H. Since the original problem requires $\dot{y} = f$, we may write this as $\dot{y} = \partial H/\partial \lambda$. This is not an additional condition, only a convenient way to express the law of motion, which together with the costate equation forms a symmetric pair.

This completes the derivation of the four conditions in the box above.

Remark. The costate $\lambda(t)$ is the *shadow price* of the state variable: it measures the marginal value, in units of the objective, of an extra unit of y at time t. The condition $\lambda(T) = 0$ when y_T is free then says that at the terminal date an extra unit of the state is worthless, since there is no time left to use it. In the Ben-Porath model this will say that human capital at the end of working life has zero value.

10.4 Other boundary conditions

The derivation above produced two boundary terms in (2), which we now read in the remaining cases.

(a) T **given,** y_T **given.** Both $\Delta T = 0$ and $\Delta y_T = 0$; no transversality condition arises. The boundary condition $y(T) = y_T$ replaces it.

(b) T **given,** y_T **free.** $\Delta T = 0$, Δy_T arbitrary: $\lambda(T) = 0$. This is the case treated above.

(c) T **free,** y_T **given.** $\Delta y_T = 0$, ΔT arbitrary: $\big[H\big]_{t=T^*} = 0$. The Hamiltonian vanishes at the optimal stopping time.

(d) T **free,** y_T **free.** Both conditions: $\lambda(T^*) = 0$ and $\big[H\big]_{t=T^*} = 0$.

(e) **Terminal curve** $y_T = \phi(T)$**.** Then $\Delta y_T = \phi'(T)\,\Delta T$, and the two boundary terms combine into $\big[H - \lambda\,\phi'(T)\big]_{t=T^*} = 0$.

Remark. Compare with the transversality conditions of the calculus of variations. There we found $\big[F - y'\partial F/\partial y'\big]_T \delta T = 0$ and $\big[\partial F/\partial y'\big]_T \delta y = 0$. The correspondence is $\lambda \leftrightarrow -\partial F/\partial y'$ and $H \leftrightarrow F - y'\partial F/\partial y'$, which the reader can verify by taking $f(t, y, u) = u$ (so that the control is simply y'). The optimal control formulation is thus a strict generalization.

10.5 Infinite horizon

When $T \to \infty$, the boundary terms in (2) become limits:

$$\lim_{t\to\infty} H(t)\,\Delta T = 0, \qquad \lim_{t\to\infty} \lambda(t)\,\Delta y_T = 0.$$

If the terminal state is free, these are usually written as

$$\lim_{t\to\infty} H(t) = 0, \qquad \lim_{t\to\infty} \lambda(t)\,y(t) = 0.$$

The second is the condition we have been using since the Ramsey model: the present value of the terminal stock of the state variable must vanish.

Remark. Some care is needed here, because the passage to the limit in (2) is not automatic. The condition $\lim \lambda(t)\,y(t) = 0$ is *not* necessary in general: Halkin (1974) [9] gives a counterexample. It *is* necessary under additional assumptions that hold in the standard growth models, in particular when the objective is discounted and the state is bounded below (Michel, 1982 [16]). More importantly for applications, the conditions of the maximum principle together with $\lim \lambda y = 0$ are *sufficient* for optimality when the Hamiltonian is concave in (y, u); this is Mangasarian's theorem, which we state at the end of the section. In practice, one takes the transversality condition as part of the characterization of the optimum, as we have done throughout.

10.6 The current-value Hamiltonian

In economics the integrand almost always has the form $F(t, y, u) = e^{-\rho t}\,G(y, u)$, with $\rho > 0$ a discount rate, and the law of motion is autonomous, $f = f(y, u)$. The explicit time dependence through $e^{-\rho t}$ makes the costate equation non-autonomous, which is inconvenient for phase-diagram analysis. The remedy is a change of variable in the costate. Define the **current-value costate**

$$\mu(t) := e^{\rho t}\,\lambda(t), \qquad \text{so that} \qquad \lambda(t) = e^{-\rho t}\,\mu(t),$$

and the **current-value Hamiltonian**

$$H^c(y, u, \mu) := G(y, u) + \mu\, f(y, u), \qquad \text{so that} \qquad H = e^{-\rho t}\,H^c.$$

We now translate the three conditions.

Control condition. $\partial H/\partial u = e^{-\rho t}\,\partial H^c/\partial u = 0$, hence $\partial H^c/\partial u = 0$.

Costate equation. Differentiating $\lambda = e^{-\rho t}\mu$ by the product rule,

$$\dot\lambda = -\rho\, e^{-\rho t}\,\mu + e^{-\rho t}\,\dot\mu = e^{-\rho t}\big(\dot\mu - \rho\,\mu\big).$$

On the other hand, $\dot\lambda = -\partial H/\partial y = -e^{-\rho t}\,\partial H^c/\partial y$. Equating the two expressions and cancelling $e^{-\rho t}$,

$$\dot\mu - \rho\,\mu = -\frac{\partial H^c}{\partial y} \quad\Longleftrightarrow\quad \dot\mu = \rho\,\mu - \frac{\partial H^c}{\partial y}.$$

Transversality. $\lim_{t\to\infty} \lambda(t)\,y(t) = \lim_{t\to\infty} e^{-\rho t}\,\mu(t)\,y(t) = 0$.

Maximum principle in current-value form

$$\frac{\partial H^c}{\partial u} = 0, \qquad \dot\mu = \rho\,\mu - \frac{\partial H^c}{\partial y}, \qquad \dot y = \frac{\partial H^c}{\partial \mu}, \qquad \lim_{t\to\infty} e^{-\rho t}\,\mu(t)\,y(t) = 0.$$

The system in (y, μ) is now autonomous. The extra term $\rho\mu$ in the costate equation has a simple reading: the shadow price in current units must grow at rate ρ net of the "dividend" $\partial H^c/\partial y$ that the state pays, exactly like an asset-pricing equation.

Example (Ramsey-Cass-Koopmans via the maximum principle)

The problem is $\max \int_0^\infty e^{-\rho t}\,U(c)\,dt$ subject to $\dot k = f(k) - c$, $k(0) = k_0$, with k_∞ free. The state is k, the control is c.

Present-value form. The Hamiltonian is

$$H(t, k, c, \lambda) = e^{-\rho t}\,U(c) + \lambda\big[f(k) - c\big].$$

The control condition gives

$$\frac{\partial H}{\partial c} = e^{-\rho t}\,U'(c) - \lambda = 0 \quad\Longrightarrow\quad \lambda = e^{-\rho t}\,U'(c).$$

The costate equation gives

$$\dot\lambda = -\frac{\partial H}{\partial k} = -\lambda\,f'(k) \quad\Longrightarrow\quad \frac{\dot\lambda}{\lambda} = -f'(k).$$

Differentiating $\lambda = e^{-\rho t}U'(c)$ with respect to t by the product and chain rules,

$$\dot\lambda = -\rho\,e^{-\rho t}\,U'(c) + e^{-\rho t}\,U''(c)\,\dot c = e^{-\rho t}\big[U''(c)\,\dot c - \rho\,U'(c)\big].$$

Substituting into $\dot\lambda = -\lambda f'(k)$ and cancelling $e^{-\rho t}$,

$$U''(c)\,\dot c - \rho\,U'(c) = -U'(c)\,f'(k) \quad\Longrightarrow\quad U''(c)\,\dot c = U'(c)\big[\rho - f'(k)\big],$$

whence

$$\dot c = -\frac{U'(c)}{U''(c)}\big[f'(k) - \rho\big],$$

the Keynes-Ramsey rule of Section 8.

Current-value form. With $\mu = e^{\rho t}\lambda$, $H^c = U(c) + \mu[f(k) - c]$ and the conditions read $\mu = U'(c)$, $\dot\mu = \rho\mu - \mu f'(k) = \mu[\rho - f'(k)]$. Differentiating $\mu = U'(c)$ gives $\dot\mu = U''(c)\,\dot c$, and substituting yields the same rule in one line. This is why the current-value form is preferred.

Transversality. The conditions are $\lim_{t\to\infty} H = 0$ and $\lim_{t\to\infty} \lambda(t)\,k(t) = 0$. The second becomes

$$\lim_{t\to\infty} e^{-\rho t}\,U'(c(t))\,k(t) = 0,$$

the condition used in Section 8. For the first, substituting $\lambda = e^{-\rho t}U'(c)$ and $f(k) - c = \dot k$,

$$H = e^{-\rho t}\big[U(c) + U'(c)\,\dot k\big],$$

so $\lim H = 0$ requires $\lim_{t\to\infty} e^{-\rho t}\big[U(c) + U'(c)\,\dot k\big] = 0$. Along the saddle path $\dot k \to 0$ and $c \to c^*$, so both terms vanish and the condition holds.

10.7 Sufficiency

The maximum principle gives necessary conditions. The following result, which we state without proof, gives a simple sufficient condition that covers all the models in this course.

Theorem 10.1 (Mangasarian, 1966 [15])**.** *Suppose F and f are continuously differentiable and concave in (y, u) for each t, and let (y^*, u^*, λ) satisfy the conditions of the maximum principle with $\lambda(t) \geq 0$ for all t whenever f is not linear in (y, u). Then (y^*, u^*) is a global maximum of the problem. In the infinite-horizon case the same holds provided in addition $\lim_{t\to\infty} \lambda(t)\big[y(t) - y^*(t)\big] \geq 0$ for every admissible y.*

In the growth models, U is concave, f is concave, and $\lambda = e^{-\rho t}U'(c) > 0$, so the hypotheses are met and the candidate found by the maximum principle is the optimum. Arrow (1968) [2] gives a weaker sufficient condition, requiring concavity in y only of the *maximized* Hamiltonian $H^0(t, y, \lambda) \coloneqq \max_u H(t, y, u, \lambda)$; it is useful when F or f fail to be jointly concave, as in some models with increasing returns.

11 Human Capital and Growth

The Ramsey model predicts long-run stagnation, and the AK model restores sustained growth at the cost of an unappealing assumption, linearity of output in physical capital. In this section we introduce **human capital** as a second accumulable factor. The idea, due to Uzawa (1965) [19] and Lucas (1988) [14], is that the knowledge embodied in people can be accumulated without bound through education, and that it is this accumulation, rather than that of physical capital, which sustains growth. We first study the problem of a single individual (Ben-Porath, 1967 [4]), which is a finite-horizon optimal control problem and illustrates the transversality condition $\lambda(T) = 0$, and then the aggregate two-sector model of Uzawa and Lucas. Further references are Barro and Sala-i-Martin (2004), ch. 5 [3], and Acemoglu (2008), ch. 10 [1].

11.1 The individual problem: Ben-Porath (1967)

Environment

An individual lives and works over $[0, T]$. At time t she holds human capital $h(t)$ and divides her time between working, a fraction $1-s(t)$, and studying, a fraction $s(t) \in [0, 1]$. The wage per unit of human capital is a constant $w > 0$, so labor income at time t is $w\,(1 - s)\,h$. Human capital evolves according to

$$\dot{h} = \phi(sh) - \delta_h\, h, \qquad h(0) = h_0 > 0,$$

where ϕ is the technology for producing human capital, with $\phi' > 0$, $\phi'' < 0$ and $\phi'(0) = +\infty$, and $\delta_h \geq 0$ is the depreciation rate (forgetting). The argument sh is the amount of human capital effectively devoted to study. The individual maximizes the present value of lifetime income, discounted at the interest rate $r > 0$:

$$\max_{s(\cdot)} \int_0^T e^{-rt}\, w\,\big(1 - s(t)\big)\, h(t)\, dt.$$

The state is h, the control is s, and $h(T)$ is free: human capital has no value after working life ends.

Hamiltonian and necessary conditions

The current-value Hamiltonian is

$$H^c(h, s, \mu) = w\,(1 - s)\,h + \mu\big[\phi(sh) - \delta_h\, h\big].$$

Control condition. Assuming an interior solution, $0 < s < 1$,

$$\frac{\partial H^c}{\partial s} = -wh + \mu\,\phi'(sh)\,h = 0.$$

Since $h > 0$, we may divide by h:

$$\mu\,\phi'(sh) = w. \tag{BP1}$$

The interpretation is that of any marginal condition: one more unit of study time costs w (forgone wage) and yields $\phi'(sh)$ units of human capital, each worth μ.

Costate equation. We compute $\partial H^c/\partial h$ by the chain rule, noting that ϕ depends on h through sh:

$$\frac{\partial H^c}{\partial h} = w\,(1-s) + \mu\big[\phi'(sh)\cdot s - \delta_h\big] = w\,(1-s) + s\,\mu\,\phi'(sh) - \mu\,\delta_h.$$

Using (BP1) in the middle term, $s\,\mu\,\phi'(sh) = s\,w$, so

$$\frac{\partial H^c}{\partial h} = w\,(1-s) + s\,w - \mu\,\delta_h = w - \mu\,\delta_h.$$

Hence

$$\dot\mu = r\mu - \frac{\partial H^c}{\partial h} = r\mu - w + \mu\,\delta_h = (r+\delta_h)\,\mu - w. \tag{BP2}$$

Transversality. T is given and $h(T)$ is free, so $\lambda(T) = 0$, that is $e^{-rT}\mu(T) = 0$, hence

$$\mu(T) = 0. \tag{BP3}$$

Solving for the costate

Equation (BP2) is a first-order linear ODE with constant coefficients, $\dot\mu - (r+\delta_h)\,\mu = -w$. A particular solution is the constant $\mu_p = w/(r+\delta_h)$, since then $\dot\mu_p = 0$ and $-(r+\delta_h)\,w/(r+\delta_h) = -w$. The general solution of the homogeneous equation is $Ce^{(r+\delta_h)t}$. Therefore

$$\mu(t) = \frac{w}{r+\delta_h} + C\,e^{(r+\delta_h)t}.$$

Imposing (BP3),

$$0 = \frac{w}{r+\delta_h} + C\,e^{(r+\delta_h)T} \quad\Longrightarrow\quad C = -\frac{w}{r+\delta_h}\,e^{-(r+\delta_h)T}.$$

Substituting back,

$$\mu(t) = \frac{w}{r+\delta_h}\Big[1 - e^{-(r+\delta_h)(T-t)}\Big]. \tag{BP4}$$

Remark. One can check directly that $\mu(t) = \int_t^T w\,e^{-(r+\delta_h)(\tau-t)}\,d\tau$. Computing the integral,

$$\int_t^T w\,e^{-(r+\delta_h)(\tau-t)}\,d\tau = w\left[\frac{-e^{-(r+\delta_h)(\tau-t)}}{r+\delta_h}\right]_{\tau=t}^{\tau=T} = \frac{w}{r+\delta_h}\Big[1 - e^{-(r+\delta_h)(T-t)}\Big],$$

which is (BP4). The shadow price of human capital at time t is therefore the present value of the wages that one additional unit of h would earn until the end of working life, discounted by interest and depreciation. It is the intertemporal version of "price of an asset equals present value of its dividends".

The life-cycle profile of investment

Substituting (BP4) into (BP1),

$$\phi'\big(s(t)\,h(t)\big) = \frac{w}{\mu(t)} = \frac{r+\delta_h}{1 - e^{-(r+\delta_h)(T-t)}}. \tag{BP5}$$

Consider the right-hand side as a function of t. The denominator $1 - e^{-(r+\delta_h)(T-t)}$ has derivative

$$\frac{d}{dt}\Big[1 - e^{-(r+\delta_h)(T-t)}\Big] = -(r+\delta_h)\,e^{-(r+\delta_h)(T-t)} < 0,$$

so the denominator is decreasing in t and the right-hand side of (BP5) is *increasing* in t. Since ϕ' is decreasing (because $\phi'' < 0$), we conclude that $s(t)\,h(t)$ is **decreasing** over the life cycle. Moreover, as $t \to T$ the denominator tends to zero and $\phi'(sh) \to +\infty$, which by the assumption $\phi'(0) = +\infty$ forces $s(t)\,h(t) \to 0$: the individual stops studying as the end of working life approaches.

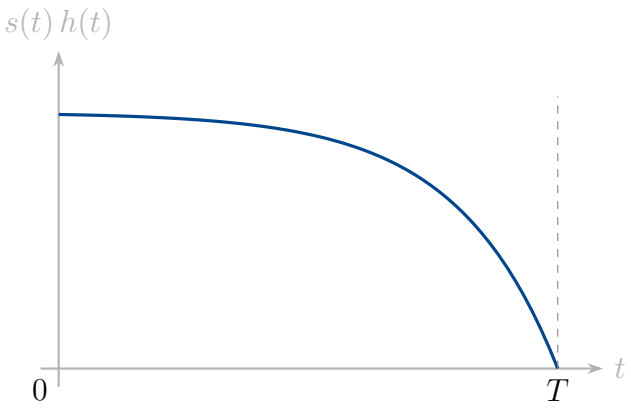


Figure 15: Investment in human capital over the life cycle in the Ben-Porath model. The effort devoted to study, $s(t)\,h(t)$, is highest early in life and tends to zero as $t \to T$, because the remaining horizon over which the returns can be collected shrinks.

Remark. In the first years of life the interior condition may fail. If $\mu(0)\,\phi'(h_0) > w$, then $\partial H^c/\partial s > 0$ at $s = 1$, and by the maximum principle (condition (i), in its full form) the optimal control is the corner $s = 1$: full-time study, no work. This is the period of *formal schooling*, followed by a phase of interior and decreasing investment (on-the-job training). The model thus reproduces the empirical pattern that schooling is concentrated at the beginning of life.

11.2 The Uzawa-Lucas model

We now turn to the aggregate economy. The Uzawa-Lucas model has two sectors: one producing the final good from physical and human capital, and an educational sector producing human capital from human capital.

Environment

A representative household has CRRA utility and discount rate $\rho > 0$. There are two stocks, physical capital $k(t)$ and human capital $h(t)$. The household allocates a fraction $u(t) \in [0, 1]$ of its human capital to production and the remaining fraction $1 - u(t)$ to education.

The final good is produced with a Cobb-Douglas technology in k and in the human capital employed in production, uh:

$$y = A\,k^{\alpha}\,(uh)^{1-\alpha}, \qquad 0 < \alpha < 1, \quad A > 0.$$

Output is consumed or invested in physical capital (we ignore depreciation):

$$\dot{k} = A\,k^{\alpha}\,(uh)^{1-\alpha} - c.$$

Human capital is produced in the educational sector with a technology that is *linear* in the human capital employed there:

$$\dot{h} = B\,(1-u)\,h, \qquad B > 0.$$

This linearity is the key assumption: it is what makes sustained growth possible, exactly as the linearity $f(k) = Ak$ did in the AK model. The difference is that the linearity now sits in the educational sector, where its interpretation (a more educated person teaches more effectively) is more palatable than linearity of goods production in physical capital.

The planner's problem is

$$\max_{c(\cdot),\,u(\cdot)} \int_0^\infty e^{-\rho t}\,\frac{c^{1-\theta}}{1-\theta}\,dt \quad \text{subject to} \quad \begin{cases} \dot{k} = A\,k^{\alpha}\,(uh)^{1-\alpha} - c, & k(0) = k_0, \\ \dot{h} = B\,(1-u)\,h, & h(0) = h_0. \end{cases}$$

There are two states (k, h) and two controls (c, u). Note that u cannot be eliminated by the trick used in the Ramsey model: it enters both laws of motion and cannot be solved for in terms of $\dot{k}$ or $\dot{h}$ alone. This is the typical situation in which the maximum principle is indispensable.

Hamiltonian and necessary conditions

With current-value costates μ (for k) and ν (for h), the Hamiltonian is

$$H^c = \frac{c^{1-\theta}}{1-\theta} + \mu\Big[A\,k^{\alpha}(uh)^{1-\alpha} - c\Big] + \nu\,B\,(1-u)\,h.$$

Control condition for c.

$$\frac{\partial H^c}{\partial c} = c^{-\theta} - \mu = 0 \quad \Longrightarrow \quad \mu = c^{-\theta}. \tag{UL1}$$

Control condition for u. The derivative of the production term is

$$\frac{\partial}{\partial u}\big[Ak^{\alpha}u^{1-\alpha}h^{1-\alpha}\big] = (1-\alpha)\,A\,k^{\alpha}\,u^{-\alpha}\,h^{1-\alpha}.$$

Hence

$$\frac{\partial H^c}{\partial u} = \mu\,(1-\alpha)\,A\,k^{\alpha}\,u^{-\alpha}\,h^{1-\alpha} - \nu\,B\,h = 0.$$

Dividing by $h > 0$ and writing $u^{-\alpha}h^{-\alpha} = (uh)^{-\alpha}$,

$$\mu\,(1-\alpha)\,A\left(\frac{k}{uh}\right)^{\alpha} = \nu\,B. \tag{UL2}$$

The interpretation: moving one unit of human capital from education to production yields $(1-\alpha)A(k/uh)^{\alpha}$ units of the final good (the marginal product of human capital in production), each worth μ, and costs B units of future human capital, each worth ν.

Costate equation for μ. The derivative of H^c with respect to k is $\mu\,\alpha A\,k^{\alpha-1}(uh)^{1-\alpha}$, so

$$\dot{\mu} = \rho\mu - \frac{\partial H^c}{\partial k} = \rho\mu - \mu\,\alpha A\,k^{\alpha-1}(uh)^{1-\alpha} = \mu\left[\rho - \alpha A\left(\frac{uh}{k}\right)^{1-\alpha}\right]. \tag{UL3}$$

Define $r \coloneqq \alpha A\,(uh/k)^{1-\alpha} = \partial y/\partial k$, the marginal product of physical capital. Then (UL3) is simply $\dot{\mu}/\mu = \rho - r$, identical to the Ramsey model.

Costate equation for ν. The derivative of H^c with respect to h has two contributions, from the production term, $\mu\,(1-\alpha)A\,k^{\alpha}\,u^{1-\alpha}\,h^{-\alpha}$, and from the education term, $\nu B(1-u)$:

$$\frac{\partial H^c}{\partial h} = \mu\,(1-\alpha)\,A\,k^{\alpha}\,u^{1-\alpha}\,h^{-\alpha} + \nu\,B\,(1-u).$$

We simplify the first term using (UL2). Writing $u^{1-\alpha}h^{-\alpha} = u \cdot u^{-\alpha}h^{-\alpha} = u\,(uh)^{-\alpha}$,

$$\mu\,(1-\alpha)\,A\,k^{\alpha}\,u^{1-\alpha}\,h^{-\alpha} = u \cdot \underbrace{\mu\,(1-\alpha)\,A\left(\frac{k}{uh}\right)^{\alpha}}_{=\nu B \text{ by (UL2)}} = \nu\,B\,u.$$

Therefore

$$\frac{\partial H^c}{\partial h} = \nu B u + \nu B(1-u) = \nu B,$$

and the costate equation becomes remarkably simple:

$$\dot{\nu} = \rho\nu - \nu B = (\rho - B)\,\nu. \tag{UL4}$$

This is the central result of the model: the return on human capital is B, a constant, independently of the levels of k and h. It is the exact analogue of the constant marginal product A in the AK model, now located in the educational sector.

Transversality. For each state,

$$\lim_{t\to\infty} e^{-\rho t}\,\mu(t)\,k(t) = 0, \qquad \lim_{t\to\infty} e^{-\rho t}\,\nu(t)\,h(t) = 0. \tag{UL5}$$

The consumption Euler equation

Differentiating (UL1) with respect to time, $\dot{\mu} = -\theta\,c^{-\theta-1}\,\dot{c}$, so

$$\frac{\dot{\mu}}{\mu} = \frac{-\theta\,c^{-\theta-1}\,\dot{c}}{c^{-\theta}} = -\theta\,\frac{\dot{c}}{c}.$$

Equating with (UL3),

$$-\theta\,\frac{\dot{c}}{c} = \rho - r \quad\Longrightarrow\quad \frac{\dot{c}}{c} = \frac{1}{\theta}\big(r-\rho\big) = \frac{1}{\theta}\left[\alpha A\left(\frac{uh}{k}\right)^{1-\alpha} - \rho\right]. \tag{UL6}$$

The form is the same as in the Ramsey and AK models. The difference lies in r: it depends on the ratio uh/k, which is endogenous and evolves over time.

The balanced growth path

We look for a trajectory along which u is constant and c, k, h grow at constant rates. We show that these rates are all equal and determine them.

Step 1: h grows at rate $B(1-u)$. Directly from the law of motion, $\dot{h}/h = B(1-u)$, which is constant if u is.

Step 2: k and c grow at the same rate as h. Dividing the law of motion of k by k,

$$\frac{\dot{k}}{k} = A\left(\frac{uh}{k}\right)^{1-\alpha} - \frac{c}{k}.$$

For $\dot{k}/k$ to be constant, both uh/k and c/k must be constant (a combination in which one rises and the other falls so as to compensate is possible only for a finite time, since each term is monotone). With u constant, uh/k constant implies $\dot{k}/k = \dot{h}/h$; and c/k constant implies $\dot{c}/c = \dot{k}/k$. Call this common rate g:

$$g = \frac{\dot{c}}{c} = \frac{\dot{k}}{k} = \frac{\dot{h}}{h} = B\,(1-u).$$

Step 3: the costates. From (UL4), $\dot{\nu}/\nu = \rho - B$. For $\dot{\mu}/\mu$, take logarithms in (UL2),

$$\ln\mu + \ln\big[(1-\alpha)A\big] + \alpha\ln k - \alpha\ln u - \alpha\ln h = \ln\nu + \ln B,$$

and differentiate with respect to time:

$$\frac{\dot{\mu}}{\mu} + \alpha\left(\frac{\dot{k}}{k} - \frac{\dot{u}}{u} - \frac{\dot{h}}{h}\right) = \frac{\dot{\nu}}{\nu}.$$

On the balanced path $\dot{u} = 0$ and $\dot{k}/k = \dot{h}/h$, so the bracket vanishes and

$$\frac{\dot{\mu}}{\mu} = \frac{\dot{\nu}}{\nu} = \rho - B.$$

Step 4: the marginal product of physical capital equals B. Comparing with (UL3), $\dot{\mu}/\mu = \rho - r$, we get

$$\rho - r = \rho - B \quad\Longrightarrow\quad r = \alpha A\left(\frac{uh}{k}\right)^{1-\alpha} = B.$$

On the balanced path the return on physical capital equals the return on human capital. This is a no-arbitrage condition between the two assets: if $r > B$ it would pay to move resources from education to production, and conversely. From this equality we obtain the long-run ratio

$$\frac{uh}{k} = \left(\frac{B}{\alpha A}\right)^{\frac{1}{1-\alpha}}.$$

Step 5: the growth rate. Substituting $r = B$ into the Euler equation (UL6),

$$g = \frac{\dot{c}}{c} = \frac{B-\rho}{\theta}.$$

Step 6: the time allocation. Since also $g = B(1-u)$, we equate:

$$B\,(1-u^*) = \frac{B-\rho}{\theta} \quad\Longrightarrow\quad 1-u^* = \frac{B-\rho}{B\theta} \quad\Longrightarrow\quad u^* = 1 - \frac{B-\rho}{B\theta} = \frac{B(\theta-1)+\rho}{B\theta}.$$

Balanced growth path of the Uzawa-Lucas model

$$g = \frac{B-\rho}{\theta}, \qquad u^* = \frac{B(\theta-1)+\rho}{B\theta}, \qquad \frac{uh}{k} = \left(\frac{B}{\alpha A}\right)^{\frac{1}{1-\alpha}}, \qquad r = B.$$

Regularity conditions

Three conditions must hold, and they turn out to coincide.

Positive growth: $g > 0 \iff B > \rho$. The productivity of the educational sector must exceed impatience.

Interior allocation: $0 < u^* < 1$. The inequality $u^* < 1$ is equivalent to $(B - \rho)/(B\theta) > 0$, that is $B > \rho$ (already imposed). The inequality $u^* > 0$ is equivalent to $B(\theta - 1) + \rho > 0$, that is

$$\rho > (1 - \theta)\, B.$$

For $\theta \geq 1$ this holds automatically; for $\theta < 1$ it is a genuine restriction.

Convergence of the objective: along the balanced path $c(t) = c(0)\, e^{gt}$ and the integrand is proportional to $e^{-\rho t} e^{(1-\theta)gt}$, which is integrable if and only if $\rho > (1 - \theta)\, g$. Substituting $g = (B - \rho)/\theta$,

$$\begin{aligned} \rho > \frac{(1-\theta)(B-\rho)}{\theta} &\iff \rho\theta > (1-\theta)B - (1-\theta)\rho \\ &\iff \rho\theta + (1-\theta)\rho > (1-\theta)B \\ &\iff \rho > (1-\theta)\, B, \end{aligned}$$

exactly the interiority condition. This is no coincidence: the objective diverges precisely when it would be optimal to devote all time to education.

Transversality: along the balanced path, μ grows at rate $\rho - B$ and k at rate g, so $e^{-\rho t}\mu k$ grows at rate

$$-\rho + (\rho - B) + g = g - B = B\,(1 - u^*) - B = -Bu^* < 0,$$

and the limit in (UL5) is zero. The same computation applies to $e^{-\rho t}\nu h$. The transversality conditions are satisfied. Note that H^c is *not* jointly concave in (k, h, c, u), because of the products $(uh)^{1-\alpha}$ and $(1 - u)h$. Writing the problem with the control $v \coloneqq uh$ (human capital employed in production), the laws of motion become $\dot{k} = Ak^{\alpha} v^{1-\alpha} - c$ and $\dot{h} = B(h - v)$, which are concave in (k, h, c, v), and the costates are positive. Mangasarian's theorem applies to this equivalent problem, so the path satisfying the conditions above is indeed optimal.

Comparison with the AK model

The formula $g = (B - \rho)/\theta$ is identical to that of the AK model with A replaced by B. The analogy is exact: in both, growth comes from a sector with a linear technology in an accumulable factor. There are, however, two important differences.

1. **Transitional dynamics.** In the AK model the economy starts immediately on the balanced path. In the Uzawa-Lucas model, if the initial ratio k_0/h_0 differs from the long-run ratio, there is an adjustment period during which u, r and the growth rates vary. Economies with abundant physical capital relative to human capital (for instance after a war that destroys people but not factories) grow differently from economies in the opposite situation. This dynamics, analyzed in detail in Barro and Sala-i-Martin (2004), ch. 5 [3], yields predictions of conditional convergence that the AK model lacks.

2. **Endogenous allocation.** In the AK model there is no intersectoral allocation decision. Here u^* is chosen optimally and responds to the parameters: greater impatience ρ or greater θ raise u^* (less time in education), while greater educational productivity B lowers u^* (more time in education). This gives the model empirical content regarding school enrollment and labor-force participation.

Remark. The technology for producing physical capital, $Ak^{\alpha}(uh)^{1-\alpha}$, has diminishing returns in each factor. Nonetheless there is sustained growth, because the factor h grows without bound and "pulls" k along. The model therefore shows that the linearity required for endogenous growth need not sit in goods production: it suffices that *some* sector has constant returns in a reproducible factor.

Remark. The original version in Lucas (1988) includes an externality: output also depends on the economy-wide *average* human capital h_a, raised to a power $\gamma > 0$, which each firm takes as given. The decentralized equilibrium then underinvests in education relative to the planner's optimum, because agents do not internalize the effect of their own human capital on the productivity of others. This is one of the classical arguments for education subsidies. The computation is left as an exercise.

12 Dynamic Programming

The methods of the previous sections, the calculus of variations and optimal control, work in continuous time and deliver the solution as a differential equation plus boundary conditions. This section develops a third approach, **dynamic programming**, which works most naturally in discrete time and delivers the solution as a *function* (the value function) that solves a functional equation (the Bellman equation). The two approaches are complementary: dynamic programming is more general (it handles uncertainty and non-smooth problems with no extra effort) and is the basis of numerical methods, while the Euler equation it produces coincides, in the smooth case, with the discrete-time analogue of what we obtained before.

The mathematical foundation of the method is the contraction mapping theorem on a complete metric space, so we begin there. The references are Kreyszig (1978), ch. 1 and 5 [11], for the functional analysis, and Stokey and Lucas (1989), ch. 3 and 4 [17], and Ljungqvist and Sargent (2018), ch. 3 [13], for the economics.

12.1 Complete metric spaces

Definition 12.1 (Metric space)**.** *A* ***metric space*** *is a set S equipped with a function $d\colon S \times S \to \mathbb{R}_+$, called a* metric*, such that for all $x, y, z \in S$:*

(i) $d(x,y) = 0$ *if and only if* $x = y$*;*

(ii) $d(x,y) = d(y,x)$*;*

(iii) $d(x,z) \le d(x,y) + d(y,z)$ *(triangle inequality).*

Definition 12.2 (Normed vector space)**.** *A* ***normed vector space*** *is a real vector space S equipped with a function $\|\cdot\| : S \to \mathbb{R}_+$, called a* norm*, such that for all $x, y \in S$ and $\alpha \in \mathbb{R}$:*

(i) $\|x\| = 0$ *if and only if* $x = 0$*;*

(ii) $\|\alpha x\| = |\alpha|\,\|x\|$*;*

(iii) $\|x + y\| \le \|x\| + \|y\|$.

Remark. Every normed space $(S, \|\cdot\|)$ is a metric space with $d(x,y) := \|x - y\|$. Indeed, (i) of the metric follows from (i) of the norm; (ii) follows from $\|x - y\| = \|(-1)(y - x)\| = |-1|\,\|y - x\| = \|y - x\|$; and (iii) follows from $\|x - z\| = \|(x - y) + (y - z)\| \le \|x - y\| + \|y - z\|$.

Definition 12.3 (Inner product)**.** *An* ***inner product*** *on a real vector space S is a function $\langle \cdot, \cdot \rangle : S \times S \to \mathbb{R}$ such that:*

(i) for each fixed $y \in S$, the map $x \mapsto \langle x, y \rangle$ is a linear functional on S;

(ii) $\langle x, y \rangle = \langle y, x \rangle$ *for all* $x, y \in S$*;*

(iii) $\langle x, x \rangle > 0$ *for all* $x \neq 0$.

The pair $(S, \langle \cdot, \cdot \rangle)$ is called an inner product space*.*

Throughout we consider vector spaces over the real numbers.

Lemma 12.4 (Cauchy-Schwarz inequality). In an inner product space, for all $x, y \in S$,

$$|\langle x,\, y\rangle| \le \|x\|\,\|y\|, \qquad \text{where } \|x\| \coloneqq \sqrt{\langle x,\, x\rangle}.$$

Proof. If $x = 0$, both sides vanish. Suppose $x \neq 0$, so that $\langle x,\, x\rangle > 0$, and define

$$\lambda \coloneqq \frac{\langle x,\, y\rangle}{\langle x,\, x\rangle}, \qquad z \coloneqq y - \lambda x.$$

We first check that z is orthogonal to x. By linearity in the first argument and symmetry,

$$\langle z,\, x\rangle = \langle y - \lambda x,\, x\rangle = \langle y,\, x\rangle - \lambda\,\langle x,\, x\rangle = \langle y,\, x\rangle - \frac{\langle x,\, y\rangle}{\langle x,\, x\rangle}\,\langle x,\, x\rangle = \langle y,\, x\rangle - \langle x,\, y\rangle = 0.$$

Now expand $\langle y,\, y\rangle$ using $y = z + \lambda x$:

$$\langle y,\, y\rangle = \langle z + \lambda x,\, z + \lambda x\rangle = \langle z,\, z\rangle + 2\lambda\,\langle z,\, x\rangle + \lambda^2\,\langle x,\, x\rangle = \|z\|^2 + \lambda^2\,\|x\|^2,$$

since $\langle z,\, x\rangle = 0$. As $\|z\|^2 \ge 0$,

$$\|y\|^2 \ge \lambda^2\,\|x\|^2 = \left(\frac{\langle x,\, y\rangle}{\|x\|^2}\right)^2 \|x\|^2 = \frac{\langle x,\, y\rangle^2}{\|x\|^2}.$$

Multiplying through by $\|x\|^2 > 0$ gives $\|x\|^2\,\|y\|^2 \ge \langle x,\, y\rangle^2$, and taking square roots, $\|x\|\,\|y\| \ge |\langle x,\, y\rangle|$. □

Proposition 12.5. *Let S be an inner product space. Then $(S, \|\cdot\|)$ with $\|x\| \coloneqq \sqrt{\langle x,\, x\rangle}$ is a normed vector space.*

Proof. We verify the three axioms. (i) $\|x\| = 0$ iff $\langle x,\, x\rangle = 0$ iff $x = 0$, by axiom (iii) of the inner product. (ii) $\|\alpha x\|^2 = \langle \alpha x,\, \alpha x\rangle = \alpha^2\,\langle x,\, x\rangle = \alpha^2\,\|x\|^2$, using linearity in each argument, so $\|\alpha x\| = |\alpha|\,\|x\|$. (iii) Expanding and using Cauchy-Schwarz,

$$\|x + y\|^2 = \langle x,\, x\rangle + 2\,\langle x,\, y\rangle + \langle y,\, y\rangle \le \|x\|^2 + 2\,\|x\|\,\|y\| + \|y\|^2 = \big(\|x\| + \|y\|\big)^2,$$

and taking square roots gives the triangle inequality. □

Definition 12.6 (Convergence, Cauchy sequences, completeness)**.** *Let (S, d) be a metric space.*

*(a) A sequence (x_n) in S **converges** to $x \in S$ if for every $\varepsilon > 0$ there exists N such that $d(x_n, x) < \varepsilon$ for all $n \ge N$. We write $x_n \to x$.*

*(b) A sequence (x_n) in S is **Cauchy** if for every $\varepsilon > 0$ there exists N such that $d(x_m, x_n) < \varepsilon$ for all $m, n \ge N$.*

*(c) The space (S, d) is **complete** if every Cauchy sequence in S converges to an element of S.*

Remark. Every convergent sequence is Cauchy: if $x_n \to x$, then given $\varepsilon > 0$ choose N with $d(x_n, x) < \varepsilon/2$ for $n \ge N$; by the triangle inequality, $d(x_m, x_n) \le d(x_m, x) + d(x, x_n) < \varepsilon$ for $m, n \ge N$. Completeness is the converse, and it is a genuine restriction, as Example 3 below shows.

Proposition 12.7 (Closed subsets of complete spaces)**.** *Let (S, d) be a complete metric space and $B \subset S$ a closed subset. Then (B, d) is a complete metric space.*

Proof. Let (x_n) be a Cauchy sequence in B. Since $B \subset S$, it is also a Cauchy sequence in S, and by completeness of S there exists $x \in S$ with $x_n \to x$. Since B is closed, it contains all its limit points, so $x \in B$. Hence every Cauchy sequence in B converges to an element of B. □

Examples

Example 1 (Euclidean space)

$\mathbb{R}^n$ with $d(x, y) = \|x - y\|$ is a complete metric space, where $\|\cdot\|$ is any norm on $\mathbb{R}^n$ (all norms on $\mathbb{R}^n$ are equivalent, so completeness does not depend on the choice). Completeness follows from the completeness of $\mathbb{R}$ applied coordinate by coordinate.

Example 2 (Banach and Hilbert spaces)

A complete normed vector space is called a **Banach space**. An inner product space $(S, \langle \cdot, \cdot \rangle)$ that is complete with respect to the norm $\|x\| = \sqrt{\langle x, x \rangle}$ is called a **Hilbert space**. Every Hilbert space is a Banach space; the converse fails.

Example 3 (a non-complete space)

Let $S = C([0,1])$ be the set of continuous real functions on $[0,1]$, with

$$d(x,y) := \int_0^1 |x(t) - y(t)|\, dt.$$

This is a metric (exercise: (i) uses that a nonnegative continuous function with zero integral is identically zero; (ii) and (iii) follow from the corresponding properties of $|\cdot|$ and monotonicity of the integral). We claim (S,d) is *not* complete.
For $n \geq 2$ define

$$x_n(t) := \begin{cases} 0, & 0 \leq t \leq \frac{1}{2}, \\ n\big(t - \frac{1}{2}\big), & \frac{1}{2} < t < \frac{1}{2} + \frac{1}{n}, \\ 1, & \frac{1}{2} + \frac{1}{n} \leq t \leq 1. \end{cases}$$

Each x_n is continuous. For $m > n$, the functions x_n and x_m coincide outside $[\frac{1}{2}, \frac{1}{2} + \frac{1}{n}]$, and on that interval both take values in $[0,1]$, so $|x_n(t) - x_m(t)| \leq 1$. Hence

$$d(x_n, x_m) = \int_{1/2}^{1/2+1/n} |x_n(t) - x_m(t)|\, dt \leq \frac{1}{n},$$

so $d(x_n, x_m) < \varepsilon$ whenever $m > n > 1/\varepsilon$. Thus (x_n) is Cauchy.
Suppose, by contradiction, that $x_n \to y$ for some $y \in S$. Let x be the step function $x(t) = 0$ for $t \leq \frac{1}{2}$ and $x(t) = 1$ for $t > \frac{1}{2}$. Then

$$\int_0^1 |x_n(t) - x(t)|\, dt = \int_{1/2}^{1/2+1/n} \Big[1 - n\big(t - \tfrac{1}{2}\big)\Big] dt = \frac{1}{n} - \frac{1}{2n} = \frac{1}{2n} \xrightarrow[n\to\infty]{} 0,$$

and, by the triangle inequality for $|\cdot|$ and monotonicity of the integral,

$$\int_0^1 |y(t) - x(t)|\, dt \leq d(y, x_n) + \int_0^1 |x_n(t) - x(t)|\, dt \xrightarrow[n\to\infty]{} 0.$$

Hence $\int_0^{1/2} |y(t)|\, dt = 0$ and $\int_{1/2}^1 |y(t) - 1|\, dt = 0$. Both integrands are continuous and nonnegative, so $y = 0$ on $[0, \frac{1}{2}]$ and $y = 1$ on $[\frac{1}{2}, 1]$, which is impossible at $t = \frac{1}{2}$. So (x_n) has no limit in S, and (S,d) is not complete.

Example 4 (the sup norm)

Let $S = C([0,1])$ with

$$d_\infty(x,y) := \sup_{t \in [0,1]} |x(t) - y(t)|.$$

The supremum is finite because a continuous function on a compact set is bounded. This is a metric: (i) if $d_\infty(x,y) = 0$ then $|x(t) - y(t)| = 0$ for every t, so $x = y$, and conversely; (ii) is immediate from $|x(t) - y(t)| = |y(t) - x(t)|$; (iii) for each t, $|x(t) - z(t)| \leq |x(t) - y(t)| + |y(t) - z(t)| \leq d_\infty(x,y) + d_\infty(y,z)$, and taking the supremum over t on the left gives the triangle inequality. Setting $\|x\|_\infty := \sup_t |x(t)|$, one checks in the same way that $\|\cdot\|_\infty$ is a norm and $d_\infty(x,y) = \|x - y\|_\infty$.

Theorem 12.8. $\big(C([0,1]), \|\cdot\|_\infty\big)$ *is a Banach space.*

Proof. Let (x_n) be a Cauchy sequence in $C([0,1])$ with respect to $\|\cdot\|_\infty$. The proof has three steps: construct a candidate limit x pointwise, show $x_n \to x$ uniformly, and show x is continuous.

Step 1: pointwise limit. Fix $t \in [0,1]$. For all n, m,

$$|x_n(t) - x_m(t)| \le \sup_{s\in[0,1]} |x_n(s) - x_m(s)| = \|x_n - x_m\|_\infty ,$$

so $(x_n(t))$ is a Cauchy sequence of real numbers. Since $\mathbb{R}$ is complete, it converges; call the limit $x(t)$. This defines a function $x\colon [0,1] \to \mathbb{R}$.

Step 2: uniform convergence. Given $\varepsilon > 0$, choose N such that $\|x_n - x_m\|_\infty < \varepsilon/2$ for all $n, m \ge N$. Fix $t \in [0,1]$ and $n \ge N$. For any $m \ge N$,

$$\begin{aligned} |x_n(t) - x(t)| &\le |x_n(t) - x_m(t)| + |x_m(t) - x(t)| \\ &\le \|x_n - x_m\|_\infty + |x_m(t) - x(t)| < \frac{\varepsilon}{2} + |x_m(t) - x(t)|. \end{aligned}$$

Since $x_m(t) \to x(t)$ as $m \to \infty$, we may choose m (depending on t) large enough that $|x_m(t) - x(t)| < \varepsilon/2$. Hence $|x_n(t) - x(t)| < \varepsilon$. As t was arbitrary and N did not depend on t,

$$\|x_n - x\|_\infty = \sup_t |x_n(t) - x(t)| \le \varepsilon \qquad \text{for all } n \ge N,$$

that is, $x_n \to x$ in $\|\cdot\|_\infty$.

Step 3: x is continuous. Fix $t \in [0,1]$ and $\varepsilon > 0$. By Step 2, choose n with $\|x_n - x\|_\infty < \varepsilon/3$. Since x_n is continuous at t, there is $\delta > 0$ such that $|x_n(t') - x_n(t)| < \varepsilon/3$ for all $t' \in [t-\delta, t+\delta] \cap [0,1]$. For such t',

$$\begin{aligned} |x(t') - x(t)| &\le |x(t') - x_n(t')| + |x_n(t') - x_n(t)| + |x_n(t) - x(t)| \\ &\le \|x - x_n\|_\infty + |x_n(t') - x_n(t)| + \|x_n - x\|_\infty \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. \end{aligned}$$

Hence x is continuous at t, and since t was arbitrary, $x \in C([0,1])$. Together with Step 2, every Cauchy sequence converges in $C([0,1])$. □

Remark. The same proof works verbatim for $C(X)$, the continuous real functions on any compact metric space X, with $\|x\|_\infty = \sup_{t\in X} |x(t)|$. This is the space we will use in dynamic programming, with $X \subset \mathbb{R}^n$ the state space. For a non-compact X one works instead with the space of *bounded* continuous functions, which is also a Banach space under the sup norm.

12.2 The contraction mapping theorem

Definition 12.9 (Contraction)**.** *Let (M,d) and (N,ρ) be metric spaces. A map $f\colon M \to N$ is a* ***contraction*** *if there exists a constant $c \in [0,1)$ such that*

$$\rho\big(f(x), f(y)\big) \le c\, d(x,y) \qquad \textit{for all } x, y \in M.$$

The constant c is called the modulus *of the contraction.*

Remark. Every contraction is uniformly continuous: given $\varepsilon > 0$, take $\delta = \varepsilon$; then $d(x,y) < \delta$ implies $\rho(f(x), f(y)) \le c\, d(x,y) \le d(x,y) < \varepsilon$, since $c < 1$. The point is that δ does not depend on x.

Theorem 12.10 (Banach fixed-point theorem)**.** *Let (M,d) be a complete metric space and $f\colon M \to M$ a contraction with modulus c. Then f has a unique fixed point $\bar{x} \in M$, that is, a unique $\bar{x}$ with $f(\bar{x}) = \bar{x}$. Moreover, for any $x_0 \in M$, the sequence of iterates $x_n \coloneqq f^n(x_0)$ converges to $\bar{x}$, and*

$$d(x_n, \bar{x}) \le \frac{c^n}{1-c}\, d(x_1, x_0).$$

Proof. **Existence.** Pick any $x_0 \in M$ and define $x_{n+1} \coloneqq f(x_n)$. We first show (x_n) is Cauchy. By the contraction property,

$$d(x_2, x_1) = d\big(f(x_1), f(x_0)\big) \le c\, d(x_1, x_0),$$

and by induction,

$$d(x_{n+1}, x_n) \le c^n\, d(x_1, x_0) \qquad \text{for all } n \in \mathbb{N}.$$

For $m > n$, applying the triangle inequality $m - n - 1$ times and then the geometric bound,

$$\begin{aligned} d(x_m, x_n) &\le d(x_m, x_{m-1}) + d(x_{m-1}, x_{m-2}) + \cdots + d(x_{n+1}, x_n) \\ &\le \big(c^{m-1} + c^{m-2} + \cdots + c^n\big)\, d(x_1, x_0) \\ &= c^n \big(c^{m-n-1} + \cdots + c + 1\big)\, d(x_1, x_0) \\ &\le c^n \sum_{j=0}^{\infty} c^j\, d(x_1, x_0) = \frac{c^n}{1-c}\, d(x_1, x_0). \end{aligned}$$

Since $0 \le c < 1$, $c^n \to 0$, so the right-hand side can be made smaller than any $\varepsilon > 0$ by taking n large. Hence (x_n) is Cauchy, and by completeness there exists $\bar{x} \in M$ with $x_n \to \bar{x}$.

$\bar{x}$ is a fixed point. By the triangle inequality and the contraction property,

$$d\big(f(\bar{x}), \bar{x}\big) \le d\big(f(\bar{x}), f(x_{n-1})\big) + d\big(x_n, \bar{x}\big) \le c\, d(\bar{x}, x_{n-1}) + d(x_n, \bar{x}),$$

where we used $x_n = f(x_{n-1})$. As $n \to \infty$, both terms on the right tend to zero because $x_n \to \bar{x}$. Hence $d(f(\bar{x}), \bar{x}) = 0$, that is, $f(\bar{x}) = \bar{x}$.

Uniqueness. Suppose $\tilde{x} \in M$ also satisfies $f(\tilde{x}) = \tilde{x}$. Then

$$d(\tilde{x}, \bar{x}) = d\big(f(\tilde{x}), f(\bar{x})\big) \le c\, d(\tilde{x}, \bar{x}),$$

so $(1-c)\, d(\tilde{x}, \bar{x}) \le 0$. Since $1 - c > 0$ and $d \ge 0$, this forces $d(\tilde{x}, \bar{x}) = 0$, hence $\tilde{x} = \bar{x}$.

Error bound. Letting $m \to \infty$ in the inequality $d(x_m, x_n) \le \frac{c^n}{1-c}\, d(x_1, x_0)$, and using continuity of d, gives $d(\bar{x}, x_n) \le \frac{c^n}{1-c}\, d(x_1, x_0)$. □

Remark. The theorem is constructive: it says not only that the fixed point exists but that iterating f from *any* starting point converges to it geometrically. This is the basis of the *value function iteration* algorithm in dynamic programming.

Application: existence and uniqueness for ODEs

Example 5 (Picard-Lindelöf)

Let $f\colon \mathbb{R} \to \mathbb{R}$ be Lipschitz, that is, there exists $c > 0$ with $|f(x) - f(y)| \le c\,|x - y|$ for all x, y. Consider the initial value problem

$$\dot{x}(t) = f\big(x(t)\big), \quad t \ge 0, \qquad x(0) = x_0.$$

Fix $\bar{t} > 0$ and let $M = C([0, \bar{t}])$ with the sup norm. Define the operator $T\colon M \to M$ by

$$(Tx)(t) := x_0 + \int_0^t f\big(x(s)\big)\, ds, \qquad t \in [0, \bar{t}].$$

A solution of the ODE on $[0, \bar{t}]$ is exactly a fixed point of T: if $Tx = x$ then $x(0) = x_0$ and, differentiating, $\dot{x} = f(x)$; conversely, integrating the ODE from 0 to t gives $x = Tx$.

T maps M into M. If x is continuous then $f \circ x$ is continuous (a Lipschitz function is continuous), so the integral is a continuous (indeed differentiable) function of t.

T is a contraction when $c\,\bar{t} < 1$. For $x, y \in M$ and $t \in [0, \bar{t}]$,

$$\begin{aligned} |(Tx)(t) - (Ty)(t)| &= \left| \int_0^t \big[f(x(s)) - f(y(s))\big]\, ds \right| \le \int_0^t |f(x(s)) - f(y(s))|\, ds \\ &\le \int_0^t c\,|x(s) - y(s)|\, ds \le c\,t\, \|x - y\|_\infty . \end{aligned}$$

Taking the supremum over $t \in [0, \bar{t}]$,

$$\|Tx - Ty\|_\infty \le c\,\bar{t}\, \|x - y\|_\infty .$$

Hence, if $c\,\bar{t} < 1$, T is a contraction on the Banach space M, and by the Banach fixed-point theorem it has a unique fixed point. This proves existence and uniqueness of the solution on $[0, \bar{t}]$. The solution on $[0, \infty)$ is obtained by repeating the argument on $[\bar{t}, 2\bar{t}]$ with initial condition $x(\bar{t})$, and so on.

12.3 Blackwell's sufficient conditions

Verifying the contraction property directly can be tedious. The following criterion, due to Blackwell (1965) [6], is often much easier to apply. We need an order on the function space.

Definition 12.11. *Let M be a space of bounded functions on a set X and $f, g \in M$. We write $f \le g$ if $f(x) \le g(x)$ for all $x \in X$. For a constant $a \in \mathbb{R}$, $f + a$ denotes the function $x \mapsto f(x) + a$.*

Theorem 12.12 (Blackwell's sufficient conditions)**.** *Let M be the space of bounded functions $X \to \mathbb{R}$ with the sup norm, or a closed subspace of it such as $C(X)$ for compact X, and let $T\colon M \to M$ satisfy:*

*(1) **(Monotonicity)** if $f, g \in M$ and $f \le g$, then $Tf \le Tg$;*

(2) ***(Discounting)*** *there exists $c \in [0,1)$ such that $T(f+a) \le Tf + c\,a$ for all $f \in M$ and all constants $a \ge 0$.*

Then T is a contraction with modulus c.

Proof. Let $f, g \in M$. For every $x \in X$, $f(x) - g(x) \le |f(x) - g(x)| \le \|f - g\|_\infty$, so

$$f \le g + \|f - g\|_\infty .$$

Applying monotonicity and then discounting with $a = \|f - g\|_\infty \ge 0$,

$$Tf \le T\big(g + \|f - g\|_\infty\big) \le Tg + c\,\|f - g\|_\infty .$$

Hence $Tf - Tg \le c\,\|f - g\|_\infty$ pointwise. Interchanging the roles of f and g,

$$Tg - Tf \le c\,\|g - f\|_\infty = c\,\|f - g\|_\infty .$$

Combining, $|Tf(x) - Tg(x)| \le c\,\|f - g\|_\infty$ for all x, and taking the supremum over x,

$$\|Tf - Tg\|_\infty \le c\,\|f - g\|_\infty .$$

□

12.4 The Bellman equation

We now turn to the economic problem. Time is discrete, $t = 0, 1, 2, \ldots$. At each date the economy is described by a *state* $x_t \in X \subset \mathbb{R}^n$, and a *control* $u_t \in \mathbb{R}^k$ is chosen, yielding a one-period return $r(x_t, u_t)$ and determining the next state through $x_{t+1} = g(x_t, u_t)$. The planner maximizes the discounted sum of returns:

$$\max_{\{u_t\}_{t \ge 0}} \sum_{t=0}^{\infty} \beta^t\, r(x_t, u_t) \qquad \text{subject to} \qquad x_{t+1} = g(x_t, u_t), \quad x_0 \text{ given},$$

with $0 < \beta < 1$.

Assumptions

We assume throughout:

(i) $r \colon X \times \mathbb{R}^k \to \mathbb{R}$ is continuous, concave and bounded;

(ii) the feasible set $\Gamma(x) \coloneqq \{x' \in X : x' = g(x,u) \text{ for some admissible } u\}$ is nonempty, compact-valued and continuous in x, and the graph $\{(x, x') : x' \in \Gamma(x)\}$ is convex;

(iii) X is compact (or, more generally, we restrict attention to bounded functions on X).

The constraint is sometimes written as $x' \le g(x,u)$, allowing free disposal; the two formulations are equivalent when r is increasing in the relevant argument, and we use the equality form for simplicity.

The principle of optimality

The key idea, due to Bellman, is that the problem has a recursive structure. Define the **value function** $v(x_0)$ as the maximized objective starting from x_0. Then the planner's choice at date 0 can be split into two parts: choose u_0 today, and from tomorrow on behave optimally starting from $x_1 = g(x_0, u_0)$, obtaining $v(x_1)$. This gives the **Bellman equation**:

Bellman equation

$$v(x) = \max_{u \in \mathbb{R}^k} \Big\{ r(x,u) + \beta\, v\big(g(x,u)\big) \Big\}, \qquad x \in X.$$

This is a *functional equation*: the unknown is the function v, which appears on both sides. The question is whether it has a solution, whether that solution is unique, and whether it coincides with the value function of the sequence problem. We answer the first two with the contraction mapping theorem; the third is the principle of optimality proper, which we take as given (see Stokey and Lucas [17], Theorems 4.2 to 4.5).

The Bellman operator

Define the operator T on $C(X)$ by

$$(Tv)(x) \coloneqq \max_{u \in \mathbb{R}^k} \Big\{ r(x,u) + \beta\, v\big(g(x,u)\big) \Big\}.$$

A solution of the Bellman equation is exactly a fixed point of T. Recall from Example 4 and the subsequent remark that $\big(C(X), \|\cdot\|_\infty\big)$ is a Banach space when X is compact.

Proposition 12.13. *Under (i) to (iii), T maps $C(X)$ into $C(X)$.*

Proof. Let $v \in C(X)$. The function $(x,u) \mapsto r(x,u) + \beta v(g(x,u))$ is continuous as a composition and sum of continuous functions. By (ii), the maximization is over a compact set that varies continuously with x. The theorem of the maximum (Berge) then guarantees that the maximum is attained and that the maximized value $(Tv)(x)$ is a continuous function of x. Boundedness of Tv follows from boundedness of r and v: $|Tv(x)| \le \sup|r| + \beta\,\|v\|_\infty$. □

Proposition 12.14. *Under (i) to (iii), T is a contraction on $\big(C(X), \|\cdot\|_\infty\big)$ with modulus β.*

Proof. We verify Blackwell's conditions.

Monotonicity. Let $v, w \in C(X)$ with $v \le w$. Then for every (x,u),

$$r(x,u) + \beta\, v\big(g(x,u)\big) \le r(x,u) + \beta\, w\big(g(x,u)\big),$$

since $\beta > 0$ and $v(g(x,u)) \le w(g(x,u))$. Taking the maximum over u on both sides preserves the inequality, so $(Tv)(x) \le (Tw)(x)$ for all x, that is, $Tv \le Tw$.

Discounting. Let $v \in C(X)$ and $a \ge 0$. Then

$$\begin{aligned} \big(T(v+a)\big)(x) &= \max_u \Big\{ r(x,u) + \beta\big[v(g(x,u)) + a\big] \Big\} \\ &= \max_u \Big\{ r(x,u) + \beta\, v(g(x,u)) \Big\} + \beta a = (Tv)(x) + \beta a, \end{aligned}$$

since the constant βa does not affect the maximization. Hence $T(v+a) = Tv + \beta a \le Tv + \beta a$, which is discounting with $c = \beta < 1$.

By Blackwell's theorem, T is a contraction with modulus β. □

Corollary 12.15. *Under (i) to (iii), the Bellman equation has a unique solution $v \in C(X)$. Moreover, for any $v_0 \in C(X)$, the iterates $v_{n+1} = Tv_n$ converge to v in the sup norm, with $\|v_n - v\|_\infty \le \frac{\beta^n}{1-\beta} \|v_1 - v_0\|_\infty$.*

Proof. Immediate from the Banach fixed-point theorem applied to T on the complete space $C(X)$. □

Remark. The convergence of iterates is *value function iteration*: start from any guess (say $v_0 = 0$), apply T repeatedly, and the result converges to the true value function. Each iteration v_n has an interpretation: it is the value function of the n-period problem with zero terminal value. The error bound shows that convergence is geometric at rate β, so a high discount factor (patient agents) makes the algorithm slow.

12.5 First-order and envelope conditions: the Euler equation

The contraction argument gives existence and uniqueness of v but says nothing about its shape. To characterize the solution, we now assume additionally that v is differentiable (this can be proved under concavity assumptions, see Benveniste and Scheinkman (1979) [5]) and that the maximum in the Bellman equation is interior.

First-order condition

Differentiating the right-hand side of the Bellman equation with respect to u and setting the result to zero,

$$\frac{\partial r}{\partial u}(x,u) + \beta\, v'\big(g(x,u)\big)\, \frac{\partial g}{\partial u}(x,u) = 0. \tag{FOC}$$

This determines the optimal control as a function of the state, $u = h(x)$, called the **policy function**.

Envelope condition

Substitute $u = h(x)$ into the Bellman equation, so that

$$v(x) = r\big(x, h(x)\big) + \beta\, v\Big(g\big(x, h(x)\big)\Big),$$

and differentiate both sides with respect to x. By the chain rule, keeping track of the dependence on x both directly and through $h(x)$,

$$v'(x) = \frac{\partial r}{\partial x} + \frac{\partial r}{\partial u}\, h'(x) + \beta\, v'\big(g(x,h(x))\big) \left[\frac{\partial g}{\partial x} + \frac{\partial g}{\partial u}\, h'(x)\right].$$

Group the terms multiplying $h'(x)$:

$$v'(x) = \frac{\partial r}{\partial x} + \beta\, v'\big(g(x,h(x))\big)\, \frac{\partial g}{\partial x} + h'(x) \underbrace{\left[\frac{\partial r}{\partial u} + \beta\, v'\big(g(x,h(x))\big)\, \frac{\partial g}{\partial u}\right]}_{=\,0 \text{ by (FOC)}}.$$

The bracket vanishes because $h(x)$ satisfies the first-order condition. This is the **envelope theorem**: the indirect effect through the optimal choice drops out. What remains is

Envelope condition (Benveniste-Scheinkman)

$$v'(x) = \frac{\partial r}{\partial x}\big(x, h(x)\big) + \beta\, v'\Big(g\big(x, h(x)\big)\Big)\, \frac{\partial g}{\partial x}\big(x, h(x)\big).$$

In the common case where the law of motion does not depend on the current state, $x' = g(u)$, we have $\partial g/\partial x = 0$ and the condition simplifies to

$$v'(x) = \frac{\partial r}{\partial x}\big(x, h(x)\big).$$

The Euler equation

Combining (FOC) and the envelope condition eliminates the unknown function v. In the case $\partial g/\partial x = 0$: the envelope condition evaluated at $x' = g(u)$ gives $v'(x') = \partial r/\partial x\,(x', u')$, and substituting into (FOC),

Euler equation

$$\frac{\partial r}{\partial u}(x, u) + \beta\, \frac{\partial r}{\partial x}(x', u')\, \frac{\partial g}{\partial u}(x, u) = 0, \qquad x' = g(u).$$

This is a second-order difference equation in the state (it involves x, x' and, through u', the state x''), the discrete-time counterpart of the Euler-Lagrange equation. The value function has disappeared; the Euler equation characterizes the optimal path directly, which is often more convenient for analytical work.

Alternative formulation: choosing next period's state

Frequently it is cleaner to treat the *next state* x' as the choice variable and to recover the control from the law of motion, $u = h(x, x')$. The Bellman equation becomes

$$v(x) = \max_{x'} \Big\{ r\big(h(x, x')\big) + \beta\, v(x') \Big\},$$

where we have written $r(u)$ for the return (depending on the state only through u, to lighten notation). The first-order condition in x' is

$$r'(u)\, \frac{\partial h}{\partial x'}(x, x') + \beta\, v'(x') = 0,$$

and the envelope condition, differentiating with respect to x with x' held at its optimum, is

$$v'(x) = r'(u)\, \frac{\partial h}{\partial x}(x, x').$$

Shifting the envelope condition one period forward, $v'(x') = r'(u')\, \partial h/\partial x\,(x', x'')$, and substituting into the first-order condition:

$$r'(u)\, \frac{\partial h}{\partial x'}(x, x') + \beta\, r'(u')\, \frac{\partial h}{\partial x}(x', x'') = 0.$$

This is the same Euler equation in different notation.

12.6 Examples

Example 6 (neoclassical growth model, $\delta = 1$)

The planner solves

$$v(k) = \max_{c} \Big\{ U(c) + \beta\, v(k') \Big\}, \qquad k' = f(k) - c,$$

where f is the production function (gross of capital, with full depreciation) and U the period utility, both strictly increasing and concave.

First-order condition in c. Since $\partial k'/\partial c = -1$,

$$U'(c) + \beta\, v'(k') \cdot (-1) = 0 \quad \Longrightarrow \quad U'(c) = \beta\, v'(k'). \tag{1}$$

Envelope condition. With $c = h(k)$ the policy function and $k' = f(k) - h(k)$, differentiating $v(k) = U(h(k)) + \beta v(f(k) - h(k))$ with respect to k,

$$v'(k) = U'(c)\, h'(k) + \beta\, v'(k') \big[f'(k) - h'(k) \big] = h'(k) \underbrace{\big[U'(c) - \beta v'(k') \big]}_{=0 \text{ by } (1)} + \beta\, v'(k')\, f'(k),$$

so

$$v'(k) = \beta\, v'(k')\, f'(k). \tag{2}$$

Combining. From (1), $v'(k') = U'(c)/\beta$, and shifting one period, $v'(k'') = U'(c')/\beta$. Evaluating (2) at k' instead of k: $v'(k') = \beta\, v'(k'')\, f'(k')$. Substituting both expressions for $v'(k')$ and $v'(k'')$,

$$\frac{U'(c)}{\beta} = \beta\, \frac{U'(c')}{\beta}\, f'(k') = U'(c')\, f'(k').$$

Multiplying by β:

$$\boxed{U'(c) = \beta\, U'(c')\, f'(k').}$$

This is the discrete-time Euler equation of the growth model: the marginal utility of consuming one unit today equals the discounted marginal utility of investing it and consuming the gross return $f'(k')$ tomorrow. Compare with the continuous-time version $\dot{c}/c = \sigma(c)[f'(k) - \rho]$ of Section 8.

Alternative route. Choosing k' directly, $v(k) = \max_{k'} \{ U(f(k) - k') + \beta v(k') \}$. The first-order condition is $-U'(c) + \beta v'(k') = 0$, and the envelope condition (now with no indirect effect, since k' is the choice) is $v'(k) = U'(c)\, f'(k)$. Shifting the latter forward and substituting into the former gives the same Euler equation in two lines. This is why the "choose next state" formulation is usually preferred.

Example 7 (cake-eating in discrete time: Euler equation and policy)

We revisit the cake-eating problem of Section 7 in discrete time. A consumer has a cake of size W_t, eats $c_t \in [0, W_t]$, and carries the rest forward, $W_{t+1} = W_t - c_t$. Preferences are $\sum_{t\geq 0} \beta^t \ln c_t$ with $0 < \beta < 1$. Choosing W' as the control, the Bellman equation is

$$v(W) = \max_{0 \leq W' \leq W} \Big\{ \ln(W - W') + \beta\, v(W') \Big\}.$$

Euler equation. The first-order condition in W' and the envelope condition in W are

$$-\frac{1}{W - W'} + \beta\, v'(W') = 0 \implies \frac{1}{c} = \beta\, v'(W'), \qquad v'(W) = \frac{1}{W - W'} = \frac{1}{c}.$$

Shifting the envelope condition forward, $v'(W') = 1/c'$, and substituting into the first-order condition,

$$\frac{1}{c} = \frac{\beta}{c'} \quad \Longrightarrow \quad c' = \beta\, c.$$

Consumption declines geometrically, the discrete analogue of $\dot{c}/c = -\rho$ from Section 7 with $\sigma = 1$.

Policy function. Since $c_t = \beta^t c_0$ and the whole cake is eventually consumed (transversality, as ln is strictly increasing),

$$W_0 = \sum_{t=0}^{\infty} c_t = c_0 \sum_{t=0}^{\infty} \beta^t = \frac{c_0}{1 - \beta} \quad \Longrightarrow \quad c_0 = (1 - \beta)\, W_0.$$

The same argument at any date gives

$$\boxed{c = h(W) = (1 - \beta)\, W, \qquad W' = \beta\, W.}$$

The consumer eats a constant fraction $1 - \beta$ of whatever cake remains.

Example 8 (cake-eating: the value function by guess-and-verify)

We now find v explicitly for the problem of Example 7. Guess $v(W) = A + B\ln W$ for constants A, B to be determined, substitute into the right-hand side of the Bellman equation, and maximize $\ln(W - W') + \beta A + \beta B \ln W'$ over W'. The first-order condition gives

$$-\frac{1}{W - W'} + \frac{\beta B}{W'} = 0 \implies W' = \frac{\beta B}{1 + \beta B} W, \qquad W - W' = \frac{W}{1 + \beta B}.$$

Substituting back and expanding the logarithms,

$$\begin{aligned}(Tv)(W) &= \ln\left(\frac{W}{1+\beta B}\right) + \beta A + \beta B \ln\left(\frac{\beta B\, W}{1+\beta B}\right) \\ &= \underbrace{\big[\beta A + \beta B \ln(\beta B) - (1+\beta B)\ln(1+\beta B)\big]}_{\text{constant}} + (1+\beta B)\ln W.\end{aligned}$$

For $v = Tv$ we match coefficients. From the coefficient of $\ln W$,

$$B = 1 + \beta B \implies B = \frac{1}{1-\beta}, \qquad \text{so } \beta B = \frac{\beta}{1-\beta}, \quad 1 + \beta B = \frac{1}{1-\beta},$$

and the policy becomes $W' = \beta W$, confirming Example 7. From the constant term, using $\ln\big(\beta/(1-\beta)\big) = \ln\beta - \ln(1-\beta)$,

$$A(1-\beta) = \frac{\beta}{1-\beta}\ln\left(\frac{\beta}{1-\beta}\right) - \frac{1}{1-\beta}\ln\left(\frac{1}{1-\beta}\right) = \frac{\beta\ln\beta + (1-\beta)\ln(1-\beta)}{1-\beta},$$

so $A = \dfrac{\ln(1-\beta)}{1-\beta} + \dfrac{\beta\ln\beta}{(1-\beta)^2}$, and therefore

$$\boxed{v(W) = \frac{\ln(1-\beta)}{1-\beta} + \frac{\beta\ln\beta}{(1-\beta)^2} + \frac{\ln W}{1-\beta}.}$$

Verification. The guess reproduced itself under T with uniquely determined constants, and T has a unique fixed point, so this is the value function. As a direct check, the lifetime utility of the policy $c_t = (1-\beta)\beta^t W$ is

$$\sum_{t=0}^{\infty} \beta^t \ln\big((1-\beta)\beta^t W\big) = \frac{\ln(1-\beta) + \ln W}{1-\beta} + \ln\beta \sum_{t=0}^{\infty} t\beta^t = \frac{\ln(1-\beta) + \ln W}{1-\beta} + \frac{\beta\ln\beta}{(1-\beta)^2},$$

using $\sum_{t\ge 0} t\beta^t = \beta/(1-\beta)^2$. This matches $v(W)$ exactly.

Remark. Example 8 is one of the few cases where the value function has a closed form. The reason is that ln is the CRRA utility with $\theta = 1$ and the technology is linear, so the problem is homothetic and v inherits the logarithmic form. The same guess-and-verify method works for the growth model with $U = \ln c$ and Cobb-Douglas $f(k) = Ak^\alpha$, with $\delta = 1$ (Exercise), giving the famous policy $k' = \alpha\beta A k^\alpha$. In general, though, v must be computed numerically by value function iteration, which is exactly what the contraction mapping theorem licenses.

Exercise 12.1. Solve the neoclassical growth model of Example 6 with $U(c) = \ln c$, $f(k) = Ak^\alpha$ and $\delta = 1$ by guess-and-verify with $v(k) = E + F \ln k$. Show that $F = \alpha/(1 - \alpha\beta)$ and that the policy function is $k' = \alpha\beta A k^\alpha$.

Exercise 12.2. Show that the Bellman operator of the cake-eating problem (Example 7) satisfies Blackwell's conditions directly, without appealing to the general proposition.

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